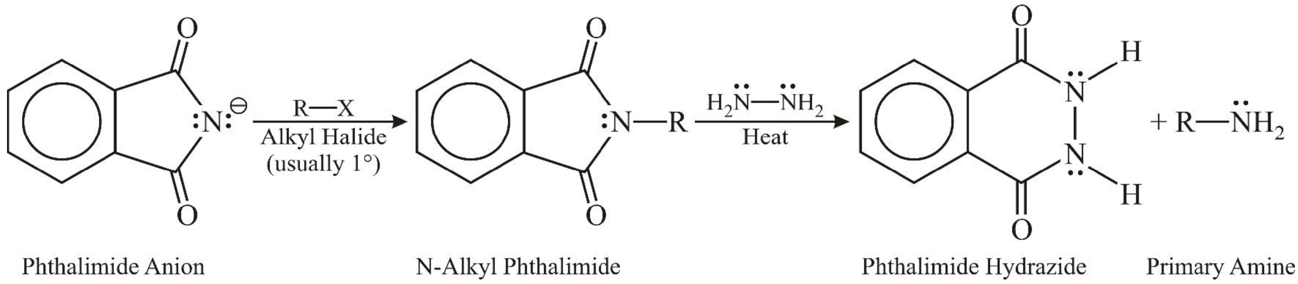
Table of Contents

BIOMOLECULES
POLYMERS

Gabriel's Phthalimide Synthesis



THEORY

1. INTRODUCTION

Complex organic compounds which govern the common activities of the living organisms are called biomolecules. Living systems are made up of various complex biomolecules like carbohydrates, proteins, nucleic acids, lipids, etc. In addition, some simple molecules like vitamins and mineral salts also play an important role in the functions of organisms.

2. CARBOHYDRATES

Carbohydrates are primarily produced by plants and form a very large group of naturally occurring organic compounds. Some common examples are cane sugar, glucose, starch etc. Most of them have a general formula, $C_xH_{2y}O_y$ and were considered as hydrates of carbon from where the name carbohydrate was derived. For example, the molecular formula of glucose ($C_6H_{12}O_6$) fits into this general formula, $C_6(H_2O)_6$. But all the compounds which fit into this formula may not be classified as carbohydrates. Rhamnose, $C_6H_{12}O_5$ is a carbohydrate but does not fit in this definition. Chemically, the carbohydrates may be defined as optically active polyhydroxy aldehydes or ketones or the compounds which produce such units on hydrolysis. Some of the carbohydrates, which are sweet in taste, are also called sugars. The most common sugar, used in our homes is named as sucrose whereas the sugar present in milk is known as lactose.

2.1 Classification of Carbohydrates

Carbohydrates are classified on the basis of their behaviour on hydrolysis. They have been broadly divided into following three groups :

2.1.1 Monosaccharides

A carbohydrate that cannot be hydrolysed further to give simpler units of polyhydroxy aldehyde or ketone is called a **monosaccharide**. Some common examples are glucose, fructose, ribose, etc.

Monosaccharides are further classified on the basis of number of carbon atoms and the functional group present in them. If a monosaccharide contains an aldehyde group, it is known as an aldose and if it contains a keto group, it is known as a ketose. Number of carbon atoms constituting the monosaccharide is also introduced in the name as is evident from the examples given

Different types of Monosaccharides

Carbon Atoms	General Term	Aldehyde	Ketone
3	Triose	Aldotriose	Ketotriose
4	Tetrose	Aldotetrose	Ketotetrose
5	Pentose	Aldopentose	Ketopentose
6	Hexose	Aldohexose	Ketohexose
7	Heptose	Aldoheptose	Ketoheptose

2.1.2 Oligosaccharides

Carbohydrates that yield two to ten monosaccharide units, on hydrolysis, are called oligosaccharides. They are further classified as disaccharides, trisaccharides, tetrasaccharides, etc., depending upon the number of monosaccharides, they provide on hydrolysis. Amongst these the most common are disaccharides. The two monosaccharide units obtained on hydrolysis of a disaccharide may be same or different. For example, sucrose on hydrolysis gives one molecule each of glucose and fructose whereas maltose gives two molecules of glucose only.

2.1.3 Polysaccharides

Carbohydrates which yield a large number of monosaccharide units on hydrolysis are called polysaccharides. Some common examples are starch, cellulose, glycogen, gums, etc. Polysaccharides are not sweet in taste, hence they are also called non-sugars.

2.1.4 Reducing and Non-Reducing Sugars

The carbohydrates may also be classified as either reducing or non-reducing sugars. All those carbohydrates which reduce Fehling's solution and Tollens' reagent are referred to as reducing sugars. **All monosaccharides whether aldose or ketose are reducing sugars.**

In disaccharides, if the reducing groups of monosaccharides i.e., aldehydic or ketonic groups are bonded, these are non-reducing sugars e.g. sucrose. On the other hand, sugars in which these functional groups are free, are called reducing sugars, for example, maltose and lactose.

3. GLUCOSE (ALDOHEXOSE)

3.1 Preparation of Glucose

(A) From Sucrose (Cane Sugar)

If sucrose is boiled with dilute HCl or H_2SO_4 in alcoholic solution, glucose and fructose are obtained in equal amounts.

$$\underset{\text{Sucrose}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{H^{\oplus}} \underset{\text{Glucose}}{C_6H_{12}O_6} + \underset{\text{Fructose}}{C_6H_{12}O_6}$$

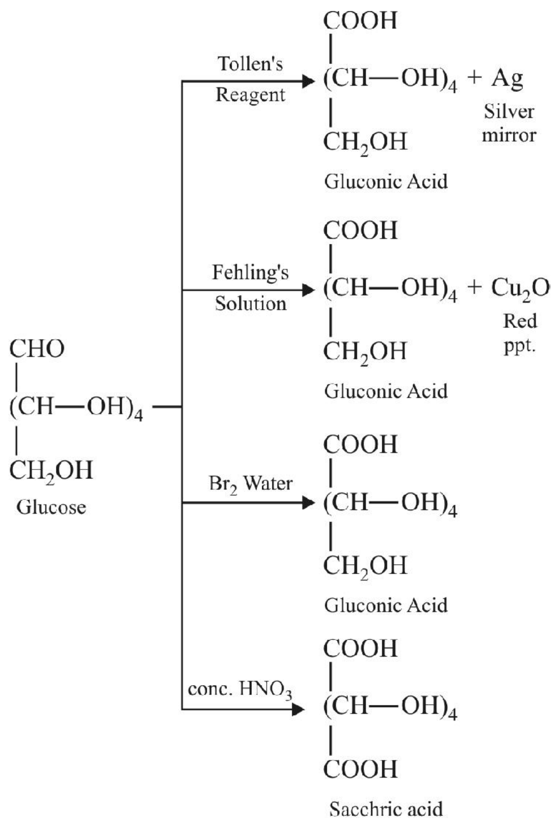
(B) From Starch

Commercially, glucose is obtained by hydrolysis of starch by boiling it with dilute H_2SO_4 at 393 K under pressure.

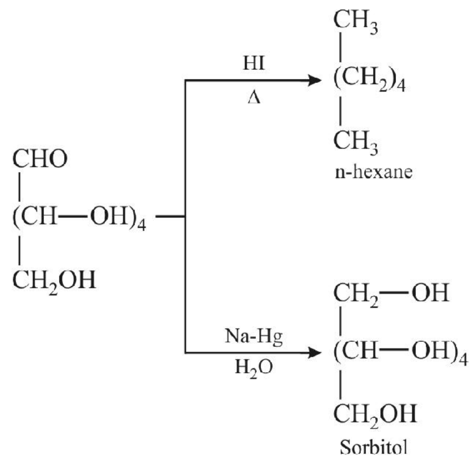
$$\underset{\text{Starch or Cellulose}}{(C_6H_{10}O_5)_n} + nH_2O \xrightarrow[393\text{ K, 2-3 atm}]{H^{\oplus}} \underset{\text{Glucose}}{nC_6H_{12}O_6}$$

3.2 Reactions

(A) Oxidation

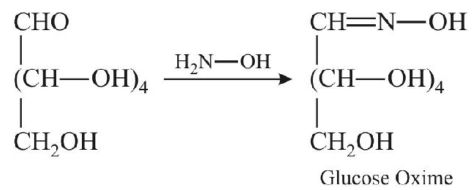


(B) Reduction

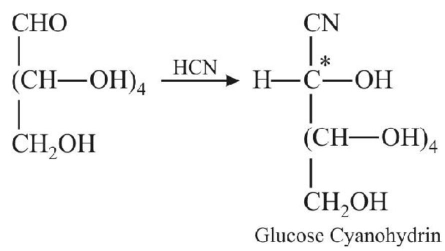


Note... Reduction with HI gives n-hexane which shows that all the 6 carbons of glucose are arranged in straight chain.

(C) Oxime Formation

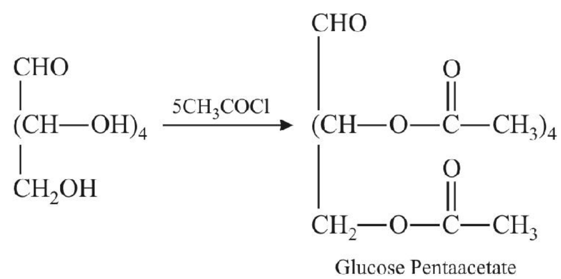


(D) Cyanohydrin Formation

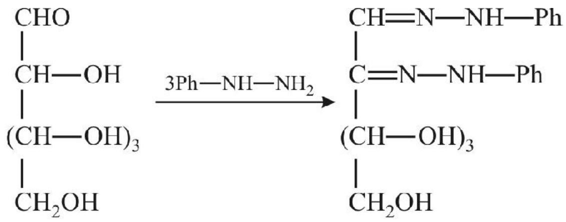


Note... Carbonyl C has become chiral so 2 products are obtained which are diastereomers.

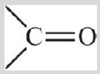
(E) Acetylation



(F) Reaction with phenylhydrazine (formation of osazone)

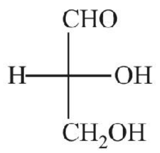


Glucosazone

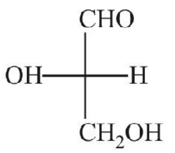
Note... One mole consumes three moles of $PhNHNH_2$ to form osazone. 2 moles give hydrazone group and one is used to oxidise CHOH group to .

3.3 Configuration in Monosaccharides

Glucose is correctly named as D(+)-glucose. 'D' before the name of glucose represents the configuration whereas '(+)' represents dextrorotatory nature of the molecule. It may be remembered that 'D' and 'L' have no relation with the optical activity of the compound. The meaning of D– and L– notations is given as follows. The letters 'D' or 'L' before the name of any compound indicate the relative configuration of a particular stereoisomer. This refers to their relation with a particular isomer of glyceraldehyde. Glyceraldehyde contains one asymmetric carbon atom and exists in two enantiomeric forms as illustrated.

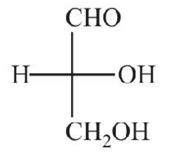


(+)-Glyceraldehyde

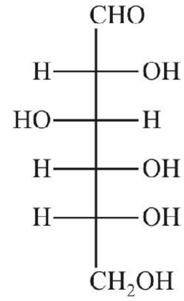


(–)-Glyceraldehyde

All those compounds which can be chemically correlated to (+) isomer of glyceraldehyde are said to have D-configuration whereas those which can be correlated to (–) isomer of glyceraldehyde are said to have L-configuration. For assigning the configuration of monosaccharides, it is the lowest asymmetric carbon atom (as shown below) which is compared. As in (+) glucose, –OH on the lowest asymmetric carbon is on the right side which is comparable to (+) glyceraldehyde, so it is assigned D-configuration. For this comparison, the structure is written in a way that most oxidised carbon is at the top.



D-(+)-Glyceraldehyde



D-(+)-Glucose

3.4 Cyclic Structure of Glucose

Glucose is found to exist in two different crystalline forms which are named as α and β. The α-form of glucose (m.p. 419 K) is obtained by crystallisation from concentrated solution of glucose at 303 K while the β-form (m.p. 423 K) is obtained by crystallisation from hot and saturated aqueous solution at 371 K.

Both α-D-glucose and β-D-glucose undergo mutarotation in aqueous solution. Although the crystalline forms of α- and β-D (+)-glucose are quite stable in aqueous solution but each form slowly changes into an equilibrium mixture of both. This is evident from the fact that the specific rotation of a freshly prepared aqueous solution of α-D(+)-glucose falls gradually from +111° to +52.5° with time and that of β-D(+)-glucose increases from +19.2° to 52.5°. Thus,

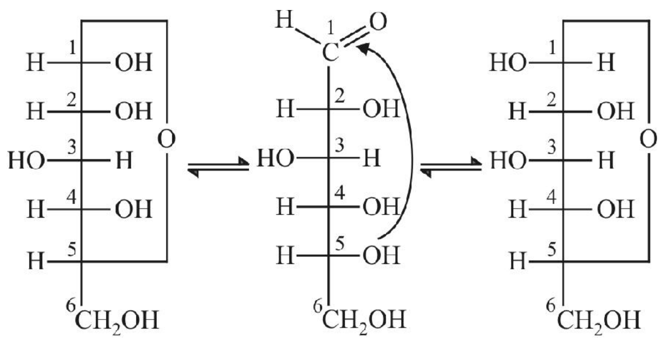
$$\underset{[\alpha]_D^{298K} = +111°}{\alpha\text{-D(+)-Glucose}} \rightleftharpoons \underset{[\alpha]_D^{298K} = +52.5°}{\text{Equilibrium mixture}}$$

$$\updownarrow$$

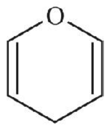
$$\underset{[\alpha]_D^{298K} = +19.2°}{\beta\text{-D(+)-Glucose}}$$

This spontaneous change in specific rotation of an optically active compound with time, to an equilibrium value, is called mutarotation.

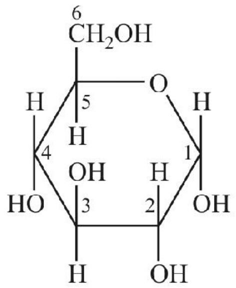
It was found that glucose forms a six-membered ring in which –OH at C-5 is involved in ring formation. This explains the absence of –CHO group and also existence of glucose in two forms as shown below. These two cyclic forms exist in equilibrium with open chain structure.



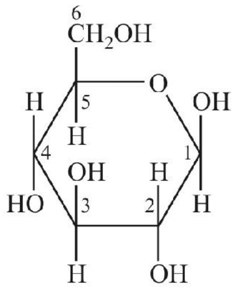
The two cyclic hemiacetal forms of glucose differ only in the configuration of the hydroxyl group at C_1, called anomeric carbon (the aldehyde carbon before cyclisation). Such isomers, i.e., α-form and β-form, are called **anomers**. The six membered cyclic structure of glucose is called **pyranose structure** (α– or β–), in analogy with pyran. Pyran is a cyclic organic compound with one oxygen atom and five carbon atoms in the ring. The cyclic structure of glucose is more correctly represented by Haworth structure.



Pyran



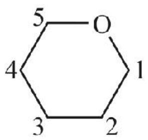
α-D-(+)-Glucopyranose



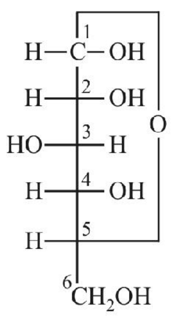
β-D-(+)-Glucopyranose

3.4.1 How to draw a Haworth Projection

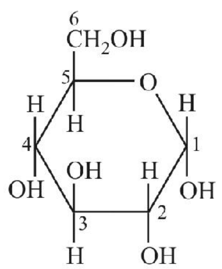
A ring of 6 atoms (5 'C' and 1 'O') is drawn in which 'O' atom is placed at right hand top corner as shown below



Carbon atom at the right hand side of oxygen is given number 1. Then other carbon atoms are given numbers 2, 3 .... in a clockwise fashion. Groups attached to a carbon in Fischer projection lying on the right hand side of that carbon are placed below the ring and on the left hand side are placed above the ring. But CH_2OH group of carbon 5 is placed above the plane of ring by convention



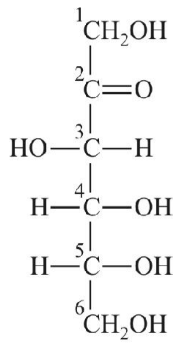
α-D-Glucose



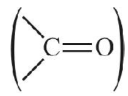
Haworth Projection of α-D-Glucose

4. FRUCTOSE (KETOHEXOSE)

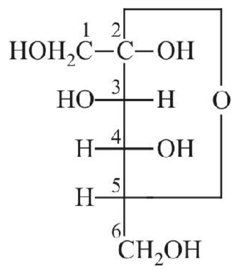
Fructose also has the molecular formula $C_6H_{12}O_6$ and on the basis of its reactions it was found to contain a ketonic functional group at carbon number 2 and six carbons in straight chain as in the case of glucose. It belongs to D-series and is a laevorotatory compound. It is appropriately written as D-(–)-fructose. Its open chain structure is as shown



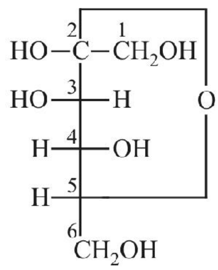
Fructose

It also exists in two cyclic forms which are obtained by the addition of –OH at C5 to the  group. The ring, thus formed is a five membered ring and is named as furanose with analogy to the compound furan. Furan is a five membered cyclic compound with one oxygen and four carbon atoms.

Furan

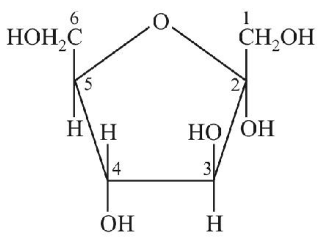


α-D-(–)-Fructofuranose

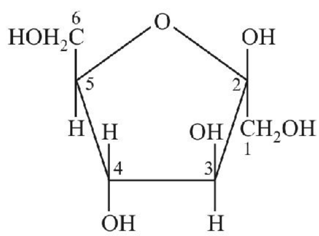


β-D-(–)-Fructofuranose

The cyclic structures of two anomers of fructose are represented by Haworth structures as given.



α-D-(–)-Fructofuranose



β-D-(–)-Fructofuranose

5. COMPARISON OF GLUCOSE AND FRUCTOSE

Comparison of Glucose and Fructose

S.No.	Property	Glucose	Fructose
1.	Molecular formula	$C_6H_{12}O_6$	$C_6H_{12}O_6$
2.	Nature	Polyhydroxy aldehyde	Polyhydroxy ketone
3.	Melting point	146°C	102°C
4.	Optical nature	Dextro rotatory	Laevo rotatory
5.	Tollen's reagent	Silver mirror	Silve mirror
6.	Fehling's solution	Red ppt	Red ppt
7.	Molisch test	Violet colour	Violet colour
8.	Phenyl hydrazine	Forms osazone	Forms osazone
9.	Oxidation by conc. HNO_3	Saccharic acid	Mixture of glycolic acid, Tartaric acid and Trihydroxy Gluteric acid

6. DISACCHARIDES

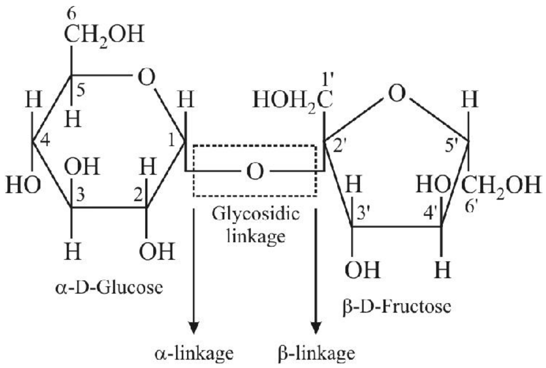
The two monosaccharides are joined together by an oxide linkage formed by the loss of a water molecule. Such a linkage between two monosaccharide units through oxygen atom is called **glycosidic linkage**.

6.1 Sucrose

One of the common disaccharides is **sucrose** which on hydrolysis gives equimolar mixture of D-(+)-glucose and D-(–)-fructose.

$$\underset{\substack{\text{Sucrose}\\ [\alpha]_D = +66.5^\circ}}{C_{12}H_{22}O_{11}} + H_2O \longrightarrow \underset{\substack{\text{D-(+)-Glucose}\\ [\alpha]_D = +52.5^\circ}}{C_6H_{12}O_6} + \underset{\substack{\text{D-(–)-Fructose}\\ [\alpha]_D = -92.4^\circ}}{C_6H_{12}O_6}$$

These two monosaccharides are held together by a glycosidic linkage between C1 of α-glucose and C2 of β-fructose. Since the reducing groups of glucose and fructose are involved in glycosidic bond formation, sucrose is a non reducing sugar.

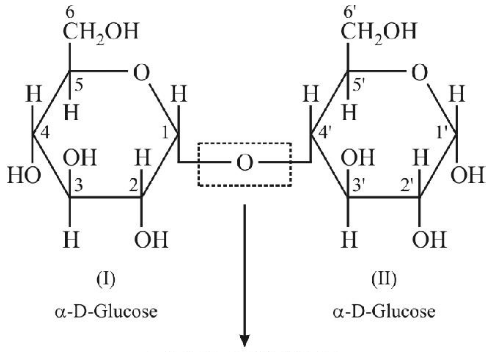


Sucrose is a dextrorotaty compound and its hydrolysis produces an equimolar solution of glucose and fructose. This solution is laevorotaty because laevo rotation of fructose is greater than dextro rotation of glucose.

Thus, hydrolysis of sucrose brings about a change in the sign of rotation, from dextro (+) to laevo (–) and the product is named as invert sugar and this phenomenon is called as inversion of sugar.

6.2 Maltose

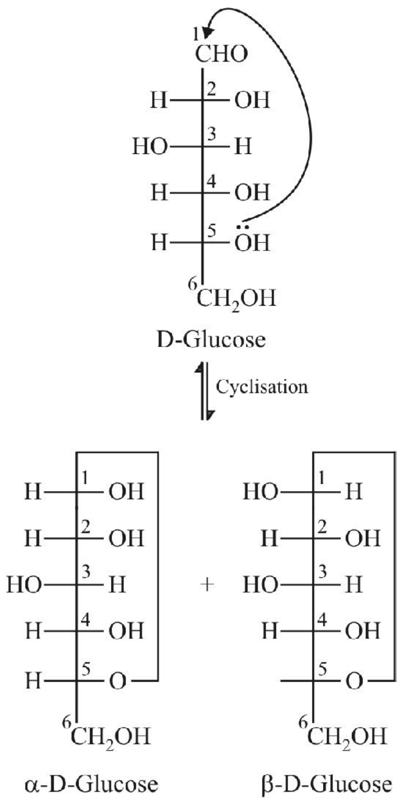
Another disaccharide, maltose is composed of two α-D-glucose units in which C1 of one glucose (I) is linked to C4 of another glucose unit (II). The free aldehyde group can be produced at C1 of second glucose in solution and it shows reducing properties so it is a reducing sugar.

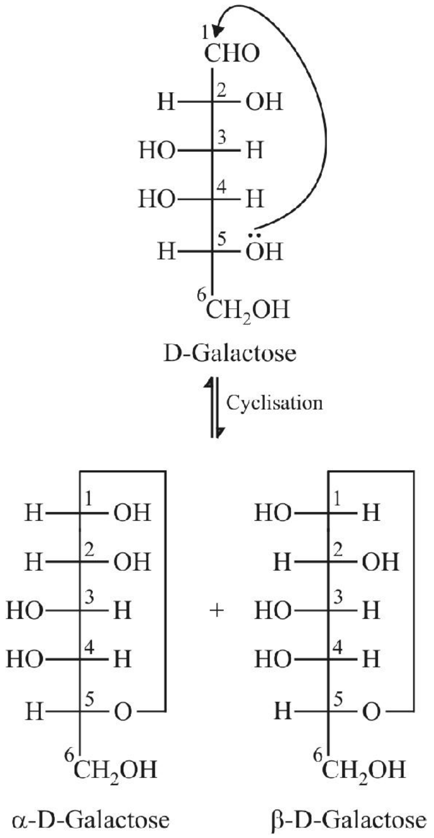


Maltose

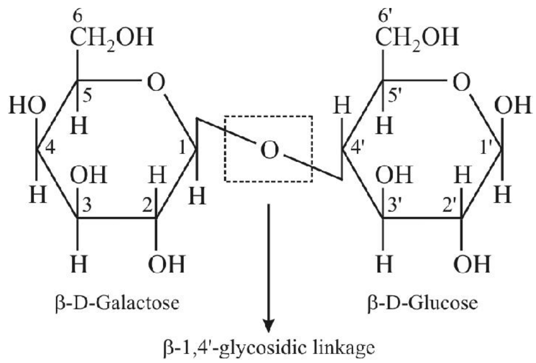
6.3 Lactose

It is more commonly known as milk sugar since this disaccharide is found in milk. It is composed of β-D-galactose and β-D-glucose. Fischer projections of β-D-Glucose and β-D-Galactose are drawn below :





We can see that the configurations of all the carbon atoms in β-D-Glucose and β-D-Galactose is same except at C-4. Such stereoisomers which differ in the configuration at only one carbon other than anomeric carbon are called as epimers and that C atom is called as epimeric carbon atom. Hence we can say that β-D-Glucose and β-D-Galactose are epimers and C-4 is epimeric carbon atom. In lactose, the linkage is between C1 of galactose and C4 of glucose. Hence it is also a reducing sugar.



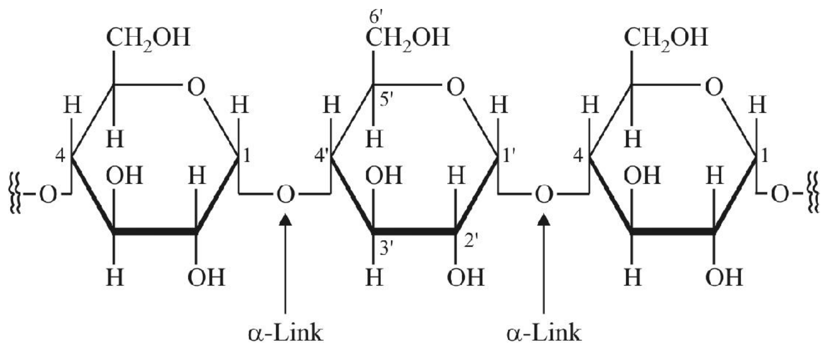
Lactose

7. POLYSACCHARIDES

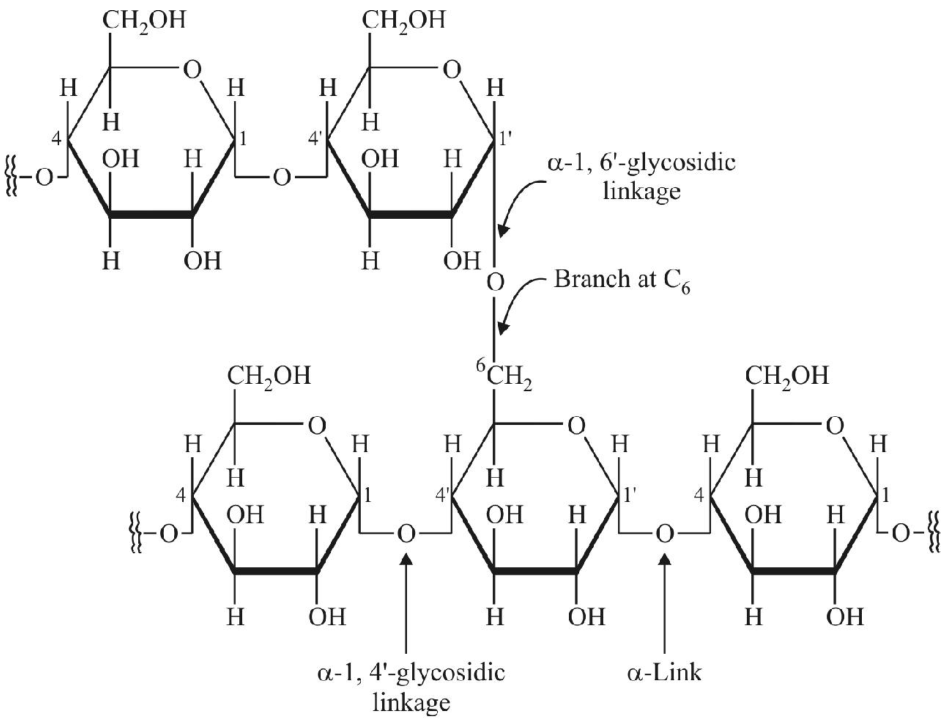
Polysaccharides contain a large number of monosaccharide units joined together by glycosidic linkages. They mainly act as the food storage or structural materials.

7.1 Starch

Starch is the main storage polysaccharide of plants. It is the most important dietary source for human beings. High content of starch is found in cereals, roots, tubers and some vegetables. It is a polymer of α-glucose and consists of two components - **Amylose** and **Amylopectin**. Amylose is water soluble component which constitutes about 15-20% of starch. Chemically amylose is a long unbranched chain with 200-1000 α-D-(+)-glucose units held by C1-C4 glycosidic linkage. Amylopectin is insoluble in water and constitutes about 80-85% of starch. It is a branched chain polymer of α-D-glucose units in which chain is formed by C1-C4 glycosidic linkage whereas branching occurs by C1-C6 glycosidic linkage.



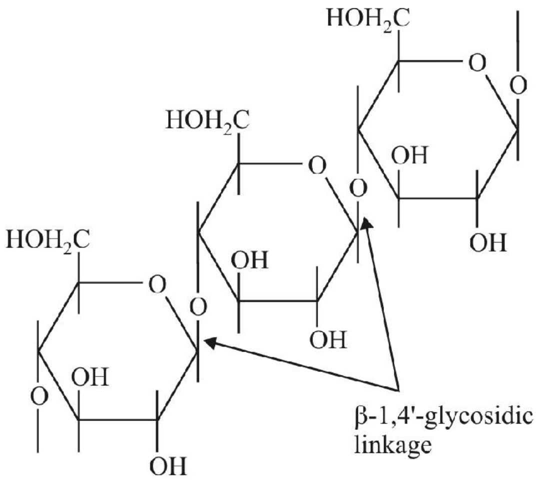
Amylose



Amylopectin

7.2 Cellulose

Cellulose occurs exclusively in plants and it is the most abundant organic substance in plant kingdom. It is a predominant constituent of cell wall of plant cells. Cellulose is a straight chain polysaccharide composed only of β-D-glucose units which are joined by glycosidic linkage between C1 of one glucose unit and C4 of the next glucose unit.



Cellulose

7.3 Glycogen

The carbohydrates are stored in animal body as glycogen. It is also known as animal starch because its structure is similar to amylopectin and is rather more highly branched. It is present in liver, muscles and brain. When the body needs glucose, enzymes break the glycogen down to glucose. Glycogen is also found in yeast and fungi.

7.4 Summary

Carbohydrates	Type	Reducing/ Non Reducing	Units	Linkage
Glucose	Monosaccharide	Reducing	Glucose	—
Fructose	Monosaccharide	Reducing	Fructose	—
Sucrose	Disaccharide	Non Reducing	α-D-Glucose and β-D-Fructose	C1 of Glucose and C2 of Fructose
Lactose	Disaccharide	Reducing	β-D-Glucose and β-D-Galactose	β-1, 4' glycosidic linkage
Maltose	Disaccharide	Reducing	Both are α-D-Glucose	α-1, 4' glycosidic linkage
Starch	Polysaccharide	Non Reducing	Amylose + Amylopectin	α-1, 4' and α-1, 6' glycosidic linkages
Glycogen	Polysaccharide	Non Reducing	Amylopectin	α-1, 4' and α-1, 6' glycosidic
Cellulose	Polysaccharide	Non Reducing	β-D-Glucose	β-1, 4' glycosidic linkage

Note...

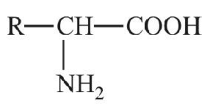
1. All the carbohydrates containing CHO group or α-Hydroxy ketonic group or hemiacetal group are reducing sugars.
2. All reducing sugars show the phenomenon of mutarotation.

8. PROTEINS

The word protein is derived from Greek word, **"proteios"** which means primary or of prime importance. All proteins are polymers of α-amino acids.

8.1 Amino Acids

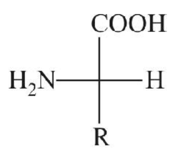
Amino acids contain amino ($-NH_2$) and carboxyl ($-COOH$) functional groups. Depending upon the relative position of amino group with respect to carboxyl group, the amino acids can be classified as α, β, γ, δ and so on. Only α-amino acids are obtained on hydrolysis of proteins. They may contain other functional groups also.

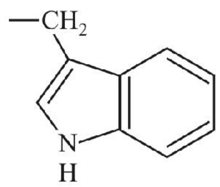
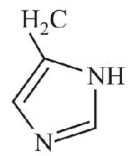
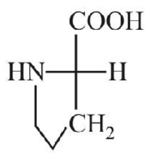


α – amino acid
(R = side chain)

All α-amino acids have trivial names, which usually reflect the property of that compound or its source. Glycine is so named since it has sweet taste (in Greek glykos means sweet) and tyrosine was first obtained from cheese (in Greek, tyros means cheese.)

8.2 Natural Amino Acids



Name of the Amino Acids	Characteristic feature of side chain. R	Three letter symbol	One letter of code
1. Glycine (N)	H	Gly	G
2. Alanine (N)	$-CH_3$	Ala	A
3. Valine* (N)	$(H_3C)_2CH-$	Val	V
4. Leucine* (N)	$(H_3C)_2CH-CH_2-$	Leu	L
5. Isoleucine* (N)	$H_3C-CH-\underset{\displaystyle CH_3}{\underset{\vert}{CH}}-$	Ile	I
6. Arginine* (B)	$HN=\underset{\displaystyle NH_2}{\underset{\vert}{C}}-NH-(CH_2)_3-$	Arg	R
7. Lysine* (B)	$H_2N-(CH_2)_4-$	Lys	K
8. Glutamic acid (A)	$HOOC-CH_2-CH_2-$	Glu	E
9. Aspartic acid (A)	$HOOC-CH_2-$	Asp	D
10. Glutamine (N)	$H_2N-\overset{\displaystyle O}{\overset{\Vert}{C}}-CH_2-CH_2-$	Gln	Q
11. Asparagine (A)	$H_2N-\overset{\displaystyle O}{\overset{\Vert}{C}}-CH_2-$	Asn	N
12. Threonine* (N)	$H_3C-CHOH-$	Thr	T
13. Serine (N)	$HO-CH_2-$	Ser	S
14. Cysteine (N)	$HS-CH_2-$	Cys	C
15. Methionine* (N)	$H_3C-S-CH_2-CH_2-$	Met	M
16. Phenylalanine* (N)	$C_6H_5-CH_2-$	Phe	F
17. Tyrosine (N)	$(p)HO-C_6H_4-CH_2-$	Tyr	Y
18. Tryptophan* (N)		Trp	W
19. Histidine* (B)		His	H
20. Proline (N)		Pro	P

N = Neutral

B = Basic

A = Acidic

Note...
1. *Essential amino acids.
2. Arginine has highest isoelectric point i.e. 10.8.
3. Cysteine has lowest isoelectric point i.e. 5.1.

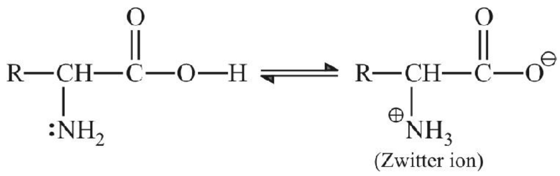
8.3 Classification of Amino Acids

Amino acids are classified as acidic, basic or neutral depending upon the relative number of amino and carboxyl groups in their molecule.

The amino acids, which can be synthesised in the body, are known as **non-essential amino acids**. On the other hand, those which cannot be synthesised in the body and must be obtained through diet, are known as **essential amino acids**.

8.4 Properties of Amino Acids

Amino acids are usually colourless, crystalline solids. These are water-soluble, high melting solids and behave like salts rather than simple amines or carboxylic acids. This behaviour is due to the presence of both acidic (carboxyl group) and basic (amino group) groups in the same molecule.



In aqueous solution, the carboxyl group can lose a proton and amino group can accept a proton, giving rise to a dipolar ion known as **zwitter ion**. This is neutral but contains both positive and negative charges. In zwitter ionic form, amino acids show amphoteric behaviour as they react both with acids and bases.

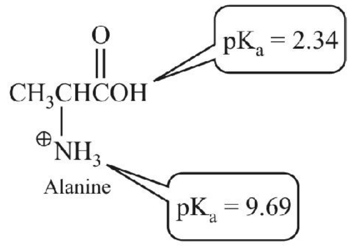
Except glycine, all other naturally occurring α-amino acids are optically active, since the α-carbon atom is asymmetric. These exist both in 'D' and 'L' forms. Most naturally occurring amino acids have L-configuration. L-Amino acids are represented by writing the $-NH_2$ group on left hand side.

8.4.1 Isoelectric Point

The **isoelectric point** (pI) of an amino acid is the pH at which it has no net charge. In other words, it is the pH at which the amount of negative charge on an amino acid exactly balances the amount of positive charge.

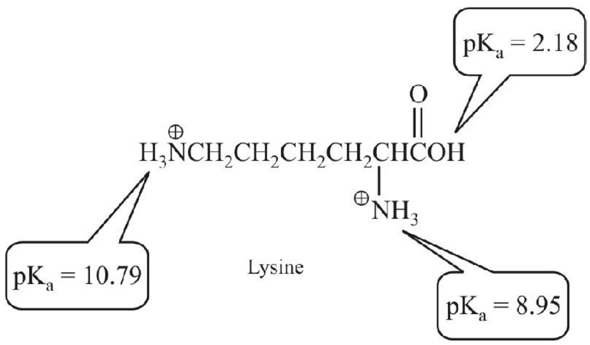
pI (isoelectric point) = pH at which there is no net charge

The pI of an amino acid that does not have an ionizable side chain–such as alanine–is midway between its two pK_a values. This is because at pH = 2.34, half the molecules have a negatively charged carboxyl group and half have an uncharged carboxyl group, and at pH = 9.69, half the molecules have a positively charged amino group and half have an uncharged amino group. As the pH increases from 2.34, the carboxyl group of more molecules becomes negatively charged; as the pH decreases from 9.69, the amino group of more molecules becomes positively charged. Therefore at the average of the two pK_a values, the number of negatively charged groups equals the number of positively charged groups.

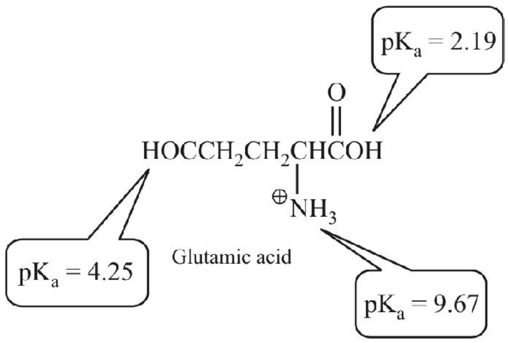


$$pI = \frac{2.34 + 9.69}{2} = \frac{12.03}{2} = 6.02$$

If an amino acid has an ionizable side chain, its pI is the average of the pK_a values of the similarly ionizing groups (positive ionizing to uncharged, or uncharged ionizing to negative). For example, the pI of lysine is the average of the pK_a values of the two groups that are positively charged in their acidic form and uncharged in their basic form. The pI of glutamate, on the other hand, is the average of the pK_a values of the two groups that are uncharged in their acidic form and negatively charged in their basic form.



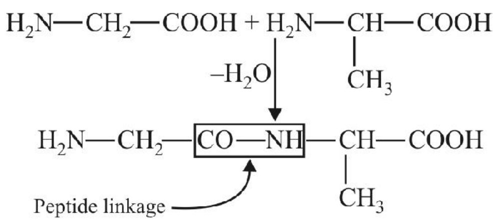
$$pI = \frac{8.95 + 10.79}{2} = \frac{19.74}{2} = 9.87$$



$$pI = \frac{2.19 + 4.25}{2} = \frac{6.44}{2} = 3.22$$

8.5 Structure of Proteins – Peptide Bond

Proteins are the polymers of α-amino acids and they are connected to each other by **peptide bond** or **peptide linkage**. Chemically, peptide linkage is an amide formed between –COOH group and $-NH_2$ group.



The reaction between two molecules of similar or different amino acids, proceeds through the combination of the amino group of one molecule with the carboxyl group of the other. This results in the elimination of a water molecule and formation of a peptide bond –CO–NH–.

8.6 Classification of Proteins

Proteins can be classified into two types on the basis of their molecular shape.

(A) Fibrous Proteins

When the polypeptide chains run parallel and are held together by hydrogen and disulphide bonds, then fibre-like structure is formed. Such proteins are generally insoluble in water.

Some common examples are **keratin** (present in hair, wool, silk) and **myosin** (present in muscles), etc.

(B) Globular Proteins

This structure results when the chains of polypeptides coil around to give a spherical shape. These are usually soluble in water. **Insulin and albumins** are the common examples of globular proteins. Structure and shape of proteins can be studied at four different levels, i.e., primary, secondary, tertiary and quaternary.

(i) Primary Structure

Proteins may have one or more polypeptide chains. Each polypeptide in a protein has amino acids linked with each other in a specific sequence and it is this sequence of amino acids that is said to be the primary structure of that protein. Any change in this primary structure i.e., the sequence of amino acids creates a different protein.

(ii) Secondary Structure

The secondary structure of protein refers to the shape in which a long polypeptide chain can exist. They are found to exist in two different types of structures viz. α-helix and β-pleated sheet structure. These structures arise due to the regular folding of the backbone of the polypeptide chain due to hydrogen bonding between C = O and –NH– groups of the peptide bond.

α-Helix is one of the most common ways in which a polypeptide chain forms all possible hydrogen bonds by twisting into a right handed screw (helix) with the –NH group of each amino acid residue hydrogen bonded to the C = O of an adjacent turn of the helix.

In β-structure all peptide chains are stretched out to nearly maximum extension and then laid side by side which are held together by intermolecular hydrogen bonds.

(iii) Tertiary Structure

The tertiary structure of proteins represents overall folding of the polypeptide chains i.e., further folding of the secondary structure. It gives rise to two major molecular shapes viz. fibrous and globular. The main forces which stabilise the 2° and 3° structures of proteins are hydrogen bonds, disulphide linkages, van der Waals and electrostatic forces of attraction.

(iv) Quaternary Structure

Some of the proteins are composed of two or more polypeptide chains referred to as sub-units. Subunits with respect to each other is known as quaternary structure.

8.7 Denaturation of Proteins

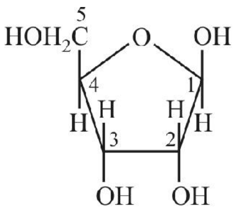
Protein found in a biological system with a unique three-dimensional structure and biological activity is called a native protein. When a protein in its native form, is subjected to physical change like change in temperature or chemical change like change in pH, the hydrogen bonds are disturbed. Due to this, globules unfold and helix get uncoiled and protein loses its biological activity. This is called **denaturation** of protein. The coagulation of egg white on boiling is a common example of denaturation. Another example is curdling of milk which is caused due to the formation of lactic acid by the bacteria present in milk.

9. NUCLEIC ACIDS

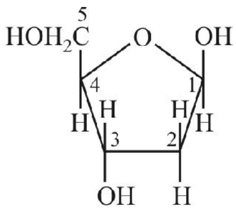
Every generation of each and every species resembles its **ancestors** in many ways. How are these characteristics transmitted from one generation to the next ? It has been observed that nucleus of a living cell is responsible for this transmission of inherent characters, also called **heredity**. The particles in nucleus of the cell, responsible for heredity, are called chromosomes which are made up of proteins and another type of biomolecules called **nucleic acids**. These are mainly of two types, the **deoxyribonucleic acid (DNA) and ribonucleic acid (RNA).** Since nucleic acids are long chain polymers of **nucleotides**, so they are also called polynucleotides.

9.1 Chemical Composition of Nucleic Acids

Complete hydrolysis of DNA (or RNA) yields a pentose sugar, phosphoric acid and nitrogen containing heterocyclic compounds (called bases). In DNA molecules, the sugar moiety is β-D-2-deoxyribose whereas in RNA molecule, it is β-D-ribose.

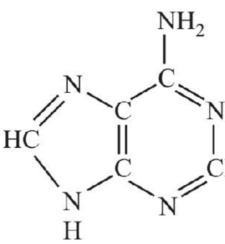


β-D-ribose

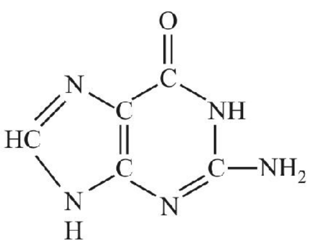


β-D-2-deoxyribose

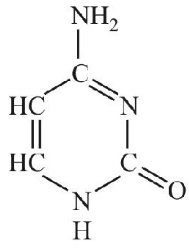
DNA contains four bases viz. **adenine** (A), **guanine** (G), **cytosine** (C) and **thymine** (T). RNA also contains four bases, the first three bases are same as in DNA but the fourth one is uracil (U).



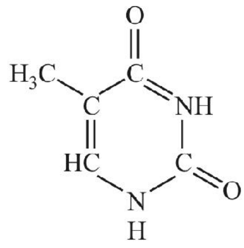
Adenine (A)



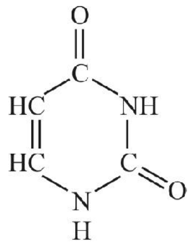
Guanine (G)



Adenine (C)



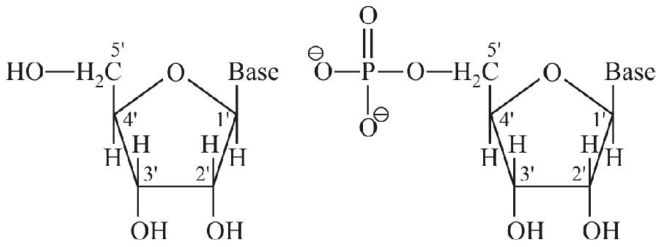
Thymine (T)



Uracil (U)

9.2 Structure of Nucleic Acids

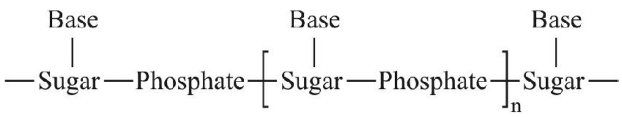
A unit formed by the attachment of a base to 1′ position of sugar is known as **nucleoside**. In nucleosides, the sugar carbons are numbered as 1′, 2′, 3′, etc. In order to distinguish these from the bases. When nucleoside is linked to phosphoric acid at 5′-position of sugar moiety, we get a nucleotide.



(a) (b)

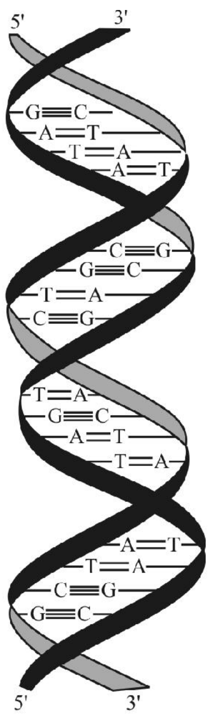
Structure of (a) a nucleoside and (b) a nucleotide

Nucleotides are joined together by phosphodiester linkage between 5′ and 3′ carbon atoms of the pentose sugar. A simplified version of nucleic acid chain is as shown below



Information regarding the sequence of nucleotides in the chain of a nucleic acid is called its primary structure. Nucleic acids have a secondary structure also. James Watson and Francis Crick gave a double strand helix structure for DNA. Two nucleic acid chains are wound about each other and held together by hydrogen bonds between pairs of bases. The two strands are complementary to each other because the hydrogen bonds are formed between specific pairs of bases. Adenine forms hydrogen bonds with thymine whereas cytosine forms hydrogen bonds with guanine.

In secondary structure of RNA, helices are present which are only single stranded. Sometimes they fold back on themselves to form a double helix structure. RNA molecules are of three types and they perform different functions. They are named as **messenger RNA (m-RNA), ribosomal RNA (r-RNA) and transfer RNA (t-RNA).**



Double strand helix structure for DNA

9.3 DNA Vs RNA

	Deoxyribonucleic Acid (DNA)	Ribonucleic Acid (RNA)
1.	DNA occurs in the nucleus of the cell.	RNA occurs in the cytoplasm of the cell.
2.	The sugar present in DNA is D-(–)-2-deoxyribose.	The sugar present in RNA is D(-)ribose.
3.	DNA contains cytosine and thymine as pyrimidine bases and guanine and adenine as purine bases.	RNA contains cytosine and uracil as pyrimidine bases and guanine and adenine as purine bases.
4.	DNA has double-stranded α-helix structure.	RNA has single stranded α-helix structure.
5.	DNA undergoes replication.	RNA usually does not undergo replication.
6.	DNA controls the transmission of hereditary effects.	RNA controls the synthesis of proteins.

9.4 Biological Functions of Nucleic Acids

DNA is the chemical basis of heredity and may be regarded as the reserve of genetic information. DNA is exclusively responsible for maintaining the identity of different species of organisms over millions of years. A DNA molecule is capable of self duplication during cell division and identical DNA strands are transferred to daughter cells. Another important function of nucleic acids is the protein synthesis in the cell. Actually, the proteins are synthesised by various RNA molecules in the cell but the message for the synthesis of a particular protein is present in DNA.

10. ENZYMES

Enzymes are biological catalysts. Chemically all enzymes are globular proteins. Some important enzymes and their functions are given

	Enzyme	Reaction Catalysed
(i)	Invertase or sucrase	Sucrose $\rightarrow$ Glucose + Fructose
(ii)	Maltase	Maltose $\rightarrow$ Glucose + Glucose
(iii)	Lactase	Lactose $\rightarrow$ Glucose + Galactose
(iv)	α-Amylase	Starch $\rightarrow$ n × Glucose
(v)	Emulsin	Cellulose $\rightarrow$ n × Glucose
(vi)	Urease	$NH_2CONH_2 \rightarrow CO_2 + 2NH_3$
(vii)	Carbonic anhydrase	$H_2CO_3 \rightarrow CO_2 + H_2O$
(viii)	Pepsin	Proteins $\rightarrow$ α-Amino acids
(ix)	Trypsin	Proteins $\rightarrow$ α-Amino acids
(x)	Nucleases	DNA or RNA $\rightarrow$ Nucleotides
(xi)	DNA polymerase	Deoxynucleotide triphosphates $\rightarrow$ DNA
(xii)	RNA polymerase	Ribonucleotide triphosphates $\rightarrow$ RNA

11. VITAMINS

	Vitamin	Sources	Deficiency diseases
(i)	Vitamin-A (Retinol or eye vitamin)	Milk, cod liver oil, butter, carrots, green leaves, tomatoes, eggs, etc.	Night blindness, xerophthalmia (i.e. hardening of cornea of eye) and xerosis.
(ii)	Vitamin-B_1 Thiamine or Aneurin	Pulses, nuts, green vegetables, Polished rice. yeast and egg yolk.	Beriberi (a disease of nervous system) and loss apetite.
(iii)	Vitamin-B_2 or Riboflavin or Lactoflavin	Milk, meat, green vegetables and yeast.	Inflammation of tongue or dark red tongue (glossitis), and cheilosis (cracking or fissuring the lips and corners of the mouth)
(iv)	Vitamin-B_6 or Pyridoxine or Adermine	Rice, bran, yeast, meat, fish, egg, yolk, maize, spinach and lettuce.	Specific dermatitlis called acrodynia, pellagra (shrivelled skin) anaemia and convulsions.
(v)	Vitamin-B_{12} or Cyanocoblamin	Milk, liver, kidney and eggs.	Inflammation of tongue, mouth etc. and pernicious anaemia.
(vi)	Vitamin-C or L-Ascorbic acid	Citrous fruits, amla, (oranges, lemons), sprouted pulses, germinated.	Scurvy and brittleness of bones, swelling and bleeding of gums and lossening of teeth.
(vii)	Vitamin D or Ergocal ciferol (or antirachtic Vitamin of sunshine vitamin)	Fish liver oil, cod liver oil, milk and eggs.	Rickets (softening and bending of bones) in children, controls Ca and P metabolism.
(viii)	Vitamin-E or Tocopherols (α, β and γ) or Antisterility factor	Eggs, milk, fish, wheat germ, oil, cotton seed oil etc.	Sterility (loss of sexual power and reproduction)
(ix)	Vitamin-H (Biotin)	Yeast, liver, Kidney and milk.	Dermatitis, loss of hair and paralysis.
(x)	Vitamin-K or Phylloquinones or Antihaemorrhagic vitamin	Cabbage, alfalfa, spinach and carrot tops.	Haemorrhage and lenthens time of blood clotting.
(xi)	Coenzyme Q_{10}	Chloroplasts of green plants and mitochondria of animals.	Low order of immunity of body against many diseases.

12. HORMONES

Hormones are biomolecules which are produced in the ductless (endocrine) glands and are carried to different parts of the body by the blood stream where they control various metabolic processes. These are required in minute quantites and unlike fats and carbohydrates these are not stored in the body but are continuously produced.

12.1 Steroidal Hormones

	Name	Organ of Secretion	Functions
A.	Sex hormones		
	(a) Androgens (Testosterone)	Testes	Control the development and normal functioning of Androsterone and male sex organs.
	(b) Estrogens (Estrone, Estradiol, Estriol)	Ovary	Control the development and normal functioning of female sex organs.
	(c) Gestogens (Progesterone)	Corpus luteum	Control the development and maintenance of pregnancy.
B.	Adrenal cortex hormones or corticoids (Cortisone, Corticosterone Aldosterone etc.)	Adrenal cortex	Regulate the metabolism of fats, proteins and carbohydrates; control the balance of water and minerals in the body.

12.2 Peptide Hormones

Name	Organ of Secretion	Functions
(i) Oxytocin	Posterior pituitary gland	Controls the contraction of the uterus after child birth and releases milk from the mammary glands.
(ii) Vasopressin	Posterior pituitary gland	Controls the reabsorption of water in the kidney.
(iii) Angiotensin II	Blood Plasma of persons	Potent vasoconstrictor i.e. contracts the blood vessels with high blood pressure.
(iv) Insulin	Pancreas	Controls the metabolism of glucose, maintains glucose level in the blood.

12.3 Amine Hormones

Name	Organ of Secretion	Functions
(i) Adrenaline or Epinephrine	Adrenal medulla	It is an amine compound and was the first hormone to be isolated. Prepares animals and humans for emergency in many ways by raising the pulse rate, blood pressure etc. stimulates the breakdown of liver glycogen into blood glucose and fats into fatty acids during emergency. These properties make adrenaline as one of the most valuable drugs used in medicine.
(ii) Thyroxine	Thyroid gland	Controls metabolism of carbohydrates, lipids and proteins.

13. TEST FOR BIOMOLECULES

13.1 Test of Carbohydrates

13.1.1 Molish Test

Molish test is used for detection of all types of carbohydrates, i.e. monosaccharides, disaccharides and polysaccharides.

Molisch reagent (1% alcoholic solution of α-naphthol) is added to the aqueous solution of a carbohydrate followed by conc. H_2SO_4 along the sides of the test tube. A violet ring is formed at the junction of the two layers.

13.2 Test for Proteins

13.2.1 Biuret Test

An alkaline solution of a protein when treated with a few drops of 1% $CuSO_4$ solution, produces a violet colouration. The colour is due to the formation of a coordination complex of Cu^{+2} with $>C=O$ and –NH– groups of the peptide linkages.

13.2.2 Xanthoproteic Test

When a protein is treated with conc. HNO_3 a yellow colour is produced. This test is given by a protein which consists of α-amino acids containing a benzene ring such as tyrosine, phenylalanine etc. and the yellow colour is due to the nitration of the benzene ring. An important example of this test is that when conc. HNO_3 is spilled on your hands, the skin turns yellow due to nitration of benzene ring of the amino acids of the proteins present in your skin.

13.2.3 Millon's Test

Millon's reagent is a solution of mercurous nitrate and mercuric nitrate in nitric acid containing little nitrous acid. When Millon's reagent is added to aqueous solution of protein, a white ppt. is formed. This test is given by all proteins containing phenolic α-amino acids i.e. tyrosine. As such gelatin which does not contain phenolic α-amino acids does not give this test.

13.2.4 Ninhydrin Test

When proteins are boiled with a dilute aqueous solution of ninhydrin (2, 2-dihydroxyindane-1,3-dione), a blue-violet colour is produced. This test is actually given by all α-amino acids. Since proteins on hydrolysis give α-amino acids, therefore, proteins and peptides also give this test.

POLYMERS

1. INTRODUCTION

A polymer may be defined as a high molecular weight compound formed by the combination of a large number of one or more types of small molecular weight compounds.

The small unit(s) of which polymer is made is (are) known as monomer.

The polymerisation may be defined as a chemical combination of a number of similar or different molecules to form a single large molecule.

A polymer which is obtained from only one type of monomer molecules is known as homopolymer. Example: polythene, PVC, PAN, teflon, Buna rubber etc.

A polymer which is obtained from more than one type of monomer is known as a co-polymer. For example: Buna-S, Dacron, Nylon-66, Bakelite etc.

2. CLASSIFICATION

2.1 Origin (Source)

2.1.1 Natural Polymers

These are of natural origin or these are found in plants and animals. Natural polymers also called as biopolymers.

Example Proteins (Polymers of amino acids), Polysaccharides (Polymers of mono saccharides), rubber (Polymers of isoprene) silk, wool, starch, cellulose, enzymes, natural rubber, haemoglobin etc.

2.1.2 Synthetic Polymers

These are artificial polymers. For example Polythene, nylon, PVC, bakelite, dacron.

2.1.3 Semi-Synthetic Polymers

Natural polymers modified according to human needs.

Examples Nitro cellulose, cellulose acetate, cellulose xanthate, etc.

2.2 Synthesis

2.2.1 Addition Polymers

These are polymers formed by the addition together of the molecules of the monomers to form a large molecule without elimination of any thing.

The process of the formation of addition polymers is called addition polymerisation.

Example–1

$$\underset{\text{Ethene}}{nCH_2{=}CH_2} \xrightarrow{\text{Polymerisation}} \underset{\text{Polythene}}{[-CH_2-CH_2-]_n}$$

2.2.2 Condensation Polymers

Condensation polymers are formed by the combination of monomers with the elimination of simple molecules such as water or alcohol. This process is called condensation polymerisation. Proteins, starch, cellulose etc. are the example of natural condensation polymers.

Two main synthetic polymers of condensation types are polyesters (Terylene or dacron) and polyamides (Nylon-66).

2.3 Mechanism

2.3.1 Chain Growth Polymerization

These polymers are formed by the successive addition of monomer units to the growing chain having a reactive intermediate (Free radical, carbocation or carbanion). Chain growth polymerisation is an important reaction of alkenes and conjugated dienes.

Polythene, polypropylene, teflon, PVC, polystyrene are some examples of chain growth polymers.

2.3.2 Step Growth Polymerization

These polymers are formed through a series of independent steps. Each step involves the condensation between two monomers leading to the formation of smaller polymer.

e.g. Nylon, terylene, bakelite etc.

2.4 Structure

2.4.1 Linear Polymers

These consist of extremely long chains of atoms and are also called one dimensional polymers. Examples - Polyethylene, PVC, Nylon, Polyester.

2.4.2 Three-Dimensional Polymers

Those polymers in which chains are cross linked to give a three dimensional network are called three dimensional polymers. Example - Bakelite.

2.5 Molecular Forces

2.5.1 Elastomers

These are the polymers having very weak intermolecular forces of attraction between polymer chains.

Elastomers posseses elastic character.

Vulcanised rubber is very important example of an elastomer.

2.5.2 Fibres

These are the polymers which have bit strong intermolecular forces such as hydrogen bonding. Ex. Nylon - 6, 6, Nylon-6, 10, Terylene.

Nylon - 6, 6 is obtained by condensation polymerisation of hexamethylene diamine (six carbon) and adipic acid (a dibasic acid having six carbon).

$$nHOOC(CH_2)_4COOH + nH_2N(CH_2)_6NH_2 \xrightarrow{\text{high pressure}} \underset{\text{Nylon-6, 6}}{[-OC(CH_2)_4CONH(CH_2)_6NH-]_n}$$

Nylon-6, 10 is obtained by condensation polymerisation of hexamethylene diamine (6C) and sebasic acid (10C).

Terylene (Dacron, teron, cronar, mylar) is a polyester fibre made by the esterification of terephthalic acid with ethylene glycol.

2.5.3 Thermoplastics

A thermo plastic polymer is one which softens on heating and becomes hard on cooling. Polyethylene, polypropylene, polystyrene are the example of thermo plastics.

2.5.4 Thermo Setting Polymers or Resin

A thermo setting polymer becomes hard on heating. Bakelite, Aniline aldehyde resin, urea formaldehyde polymer.

3. MONOMERS AND POLYMERS

S.N.	Monomer	Polymer	Type of Polymers
1.	$CH_2{=}CH_2$ (Ethylene)	Poly ethene	Addition polymer
2.	$CH_2{=}CHCH_3$ (Propylene)	Poly propylene or koylene	Addition homo polymer
3.	$CH_2{=}CHCl$ (Vinyl chloride)	Polyvinyl chloride (PVC)	Homopolymer, chain growth
4.	$CH_2{=}CH{-}C_6H_5$ (Styrene)	Polystyrene (styron)	Addition homo polymer, linear chain
5.	$CH_2{=}CH{-}CN$ (Acrylonitrile)	Polyacrylonitrile (PAN) or Orlon	Addition homopolymer
6.	$CH_2{=}CH{-}CH{=}CH_2$ (1,3 Butadiene)	BUNA rubbers	Addition copolymer
7.	$CH_2{=}CHOCOCH_3$ (Vinyl acetate)	Poly vinyl acetate (PVA)	Addition homopolymer
8.	$CF_2{=}CF_2$ (Tetrafluoro ethylene)	Teflon	Chain growth homopolymer (Nonstick cookwares)
9.	$CH_2{=}C(CH_3){-}CH{=}CH_2$ (Isoprene)	Natural Rubber	Addition homopolymer
10.	$CH_2{=}C(Cl){-}CH{=}CH_2$ (Chloroprene)	Neoprene (Artificial Rubber)	Addition homopolymer
11.	Ethylene Glycol + dimethyl terephthalate	Terylene or Dacron (Polyester)	Copolymer, step growth
12.	Hexamethylene diamine + adipic acid	Nylon-6,6 (Polyamide)	Copolymer, step growth linear
13.	Formaldehyde + urea	Urea formaldehyde resin	Copolymer, step growth
14.	Formaldehyde + Phenol	Bakelite	Copolymer, step growth thermo setting polymer
15.	Maleic anhydride + methylene glycol	Alkyl plastic	
16.	Methyl methacrylate	Poly methyl meth acrylate (PMMA)	Addition homopolymer
17.	Ethylene Glycol + Phthalic acid	Glyptal	Copolymer, linear step growth, thermo plastic
18.	Melamine + formaldehyde	Melamine formaldehyde resin	Copolymer, step growth thermosetting polymer
19.	Hexamethylene diamine + sebasic acid	Nylon - 6,10	Copolymer, step growth linear
20.	6-Aminohexanoic acid	Nylon - 6	Homopolymer, step growth linear

CHEMISTRY IN EVERYDAY LIFE

1. MEDICINE OR DRUGS

Chemical substances helping to a human body or an animal either for treatment of diseases or to reduce suffering from pain are called medicine or drug.

The treatment of disease by chemical compound which destroy the micro organism without attacking the tissue of the human body is known as chemotherapy, and the compounds used are called chemotherapeutic agent.

Various type of medicinal compounds are.

1.1 Antiseptics

Which prevent or destroy the growth of the harmful micro organism, common antiseptics are - Dettol, Savlon, Cetavelon, acriflavin, iodine, methylene blue, mercurochrome & $KMnO_4$.

Dettol is a mixture of chloroxylenol and terpineol. Its dilute solution is used to clean wounds.

Bithional - It is added to soap to impart antiseptic properties.

1.2 Disinfectants

The chemical compounds capable of completely destroying the micro organism are termed as disinfectants. These are toxic to living tissues.

These are utilized for sterilization of floor, toilets instruments & cloths.

eg. 1% solution of phenol is disinfectant while 0.2% solution of phenol is antiseptic.

1.3 Analgesics

The substances which are used to get relief from pain. These are of two types :

(a) Narcotics or habit forming drugs

(b) Non-narcotics

(a) Narcotics : These are alkaloids and mostly opium products, causes sleep and unconciousness when taken in higher doses.

e.g. Morphine, codeine, heroine

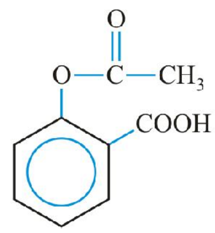
(b) Non-narcotics : Analgesics belonging to this category are effective antipyretics also.

e.g. Aspirin & novalgin, Ibuprofen, Naproxen.

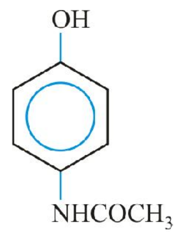
1.4 Antipyretics

To bring down the body temp. in high fever are called antipyretics.

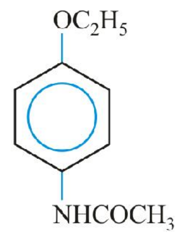
Example–1



Aspirin



Paracetamol



Phenacetin

1.5 Antimalarials

To bring down the body temperature during malarial fever.

e.g. Quinine, Chloroquine, Paraquine and Primaquine etc.

1.6 Tranquilizers

The chemical substances which act on the central nervous system and has a calming effect. Since these are used for mental diseases so are known as psycotherapeutic drugs.

They are of two types -

(a) Sedative or hypnotics (b) Mood elevators

(a) Sedative : Reduce nervous tension and promote relaxation. e.g. Reserpine, barbituric acid and its derivatives as luminal & seconal.

(b) Mood elevators or Antidepressants : A drug used for treatment of highly depressed patient, who has lost his confidence.

e.g. Benzedrine (amphetamine)

1.7 Anaesthetics

These are chemical substances helping for producing general or local insensibility to pain and other sensation. These are of two types (a) General (b) Local

(a) General - Produce unconciousness and are given at the time of major surgical operations.

e.g. Gaseous form – Nitrous oxide, ethylene, cyclopropane etc.

Liquid form – Chloroform, divinyl ether and sodium pentothal etc.

(b) Local anaesthetics : Produce loss of sensation on a small portion of the body. It is used for minor operations.

e.g. Jelly form – Oxylocain

Spray form – Ethyl chloride

Injection form – Procain

1.8 Antibiotics

The chemical substances produced from some micro organism (fungi, bacteria or mold) and are used to inhibit the growth of other micro organism.

These are effective in the treatment of infections diseases.

e.g. Penicillin - It is highly effective drug for pneumonia, Bronchitis, abcesses, sore throat etc.

Synthetic antibiotics are Streptomycin - (Tuberculosis),

Chloromycetin - (Typhoid, Meningitis, Pneumonia, diarrhoea, dysentary etc.)

Tetracyclins - (Acute fever, trachoma, dysentery & urinary tract infection)

1.9 Sulpha Drugs

Having great antibacterial powers. These are a group of drugs which are derivatives of sulphanilamide.

Other sulpha drugs are -

(a) Sulphathiazole - Mainly used in severe infections.

(b) Sulpha guanidine - Used in bacillary dysentery

(c) Sulpha pyridine - Used in pneumonia

(d) Sulpha diazine - Used in dysentery, urinary infection and respiratory infection.

2. ROCKET PROPELLANTS

2.1 Introduction

In order to provide sufficient push to the rocket satellites to enter into the space, some chemical fuels are used, which are termed as rocket propellants.

A propellant is a combination of two compounds i.e.

(a) An explosive compound called fuel

(b) Oxidiser

A chemical compound should satisfy the following conditions to function as propellant -

1. The burning of fuel should not leave any ash.
2. The burning of fuel should produce a large volume of gases/g of fuel.
3. The combustion should proceed at a fast rate.

2.2 Classification

Depending upon physical state of fuel and oxidiser, the propellants are of three types

(a) Solid propellants

(b) Liquid propellants

(c) Hybrid propellants

2.2.1 Solid Propellants

In which fuel and oxidiser both are solid. These are of two types

(I) Composite propellant : It contains polymeric binder as fuel and ammonium perchlorate as oxidiser.

Fuel - Polyurethane or polybutadiene,

Oxidiser - Ammonium perchlorate

(II) Double base propellant : It consists of nitro cellulose and nitroglycerine

Disadvantage of solid propellant

Once they ignite, they burn with a predetermined rate.

These do not have the start and stop capability.

2.2.2 Liquid Propellants

These are of two types :

(I) Monoliquid propellant : when a single liquid acts as fuel and oxidiser.

eg. Nitromethane, Methyl nitrate, H_2O_2 etc.

(II) Biliquid propellant : It comprises a liquid fuel and a liquid oxidiser e.g. Fuel – Kerosene, alcohol, hydrazine, monomethyl hydrazine (MMH) or liquid hydrogen.

Oxidiser – Liquid oxygen, nitrogen tetraoxide (N_2O_4) or nitrous acid.

Advantages :

(I) These provide higher thrust than solid propellants.

(II) The thrust can be controlled by switching on and off the flow of liquid propellant.

2.2.3 Hybrid Propella nts

These consist of a solid fuel and a liquid oxidizer. e.g.

Fuel – Acrylic rubber

Oxidiser – Liquid N_2O_4

2.3 Specific Impulse (I_s)

The superiority and performance of a propellant is expressed in terms of specific impulse (I_s).

$$I_s = \sqrt{\frac{T}{M}}$$

Where T = Flame temperature, M = average molecular mass.

Thus the performance of rocket propellant will be better if flame temperature is higher and the average mass of the product gas is lower.

3. DYES

3.1 Introduction

Dyes are coloured substances which can be applied in solution or dispersion to a substrate such as textile fibres (cotton, wool, silk, polyester, nylon), paper, leather, hair, fur, plastic material etc. giving it a coloured appearance.

If a compound absorbs light in the visible region, its colour will be that of the reflected light, i.e. complementary to that absorbed. For example, if a dye absorbs blue colour, it will appear yellow which is the complementary colour of blue.

Auxochromes are groups which themselves do not absorb light (i.e. are not chromophores) but deepen the colour when introduced into the coloured compounds, i.e., OH, NH_2, Cl, CO_2H etc.

3.2 Classification

3.2.1 Source

(i) Natural dyes are obtained from plants. For example, alizarin, indigo etc.

(ii) Synthetic dyes are prepared in the laboratory. For example, martius yellow, malachite green, orange-I, orange-II, congo red, aniline yellow etc.

3.2.2 Constitution

(i) Nitro dyes : martius yellow

(ii) Azo dyes : aniline yellow, methyl orange, orange-I, congo red etc.

(iii) Triphenylmethane dyes : malachite green, magneta

(iv) Indigoid dyes : Indigo, indigosol

(v) Anthraquinone dyes : alizarin

(vi) Phthalein dyes : phenolphthalein

3.2.3 Application

(A) Acid Dyes

Acid dyes are sodium salts of azo dyes containing sulphonic acid or carboxylic acid groups, e.g. orange-I, orange-II, congo red, methyl orange and methyl red. These dyes do not have any affinity for cotton but are used to dye wool, silk, polyurethane fibres. The affinity of acid dyes for nylon is higher than that for other types because polycaprolactam fibres contain a higher proportion of free amino groups.

(B) Basic Dyes

Basic dyes are the salts of azo and triphenylmethane dyes containing amino groups as auxochromes, e.g., aniline yellow, butter yellow, malachite green and chrysoidine G. These dyes are applied in their soluble acid solutions and get attached to the anionic sites present on the fabrics. Such dyes are used to dye polyesters and reinforced nylons.

(C) Direct Dyes

Direct dyes are water soluble dyes which are directly applied to the fabric from an aqueous solution. These dyes are most useful for fabrics which can form H-bonds such as cotton, wool, silk, rayon and nylon. Some example of direct dyes are : congo red and martius yellow.

(D) Fibre Reactive

Fibre reactive dyes attach themselves to the fibre by an irreversible chemical reaction. Therefore, their dyeing is fast and colour is retained for a longtime. These dyes contain a reactive group which combines directly with the hydroxyl or amino group of the fibre (cotton, wool, silk). Dyes which are derivatives of 2, 4-dichloro-1, 3, 5-triazine are important examples of fibre reactive dyes.

(E) Insoluble Azo Dyes

Insoluble azo dyes constitute about 60% of the total dyes used. These are obtained by coupling of phenols, naphthols, aminophenols adsorbed on the surface of a fabric with a polyurethanes, polyacrylonitriles and leather. Azo dyes can also be used to dye foodstuffs, cosmetics, drugs, biological strains such as indicators etc. However, because of their toxic nature, these dyes are no longer permitted to dye foodstuffs.

(F) Ingrain Dyes

Ingrain dyes are water insoluble azo dyes which are produced in situ on the surface of the fabric by means of coupling reactions. e.g. para red.

(G) Vat Dyes

Vat dyes are insoluble dyes which are first reduced to colourless soluble form (leuco compound) in large vats with a reducing agent such as alkaline sodium hydrosulphite and applied to the fabric and then oxidised to the insoluble coloured form by exposure to air or some oxidising agent such as chromic acid or perboric acid, e.g., indigo. Indigosol O, on the other hand, is readily fibre with formation of indigo. It is especially suitable for wool.

(H) Mordant Dyes

Mordant dyes are applied to the fabric after treating them with a metal ion (mordant) which acts as a binding agent between the dye and the fabric. Depending upon the metal ion used, the same dye can give different colours. Thus, alizarin gives a rose red (turkey red) colour with Al^{+3} ions and blue colour with Ba^{2+} ions. These dyes are especially used for dyeing wool.

4. CHEMICALS IN FOOD

4.1 Food additives

All those chemicals which are added to food to improve its keeping qualities, appearance, taste, odour and nutritive or food value are called food additives.

These include preservatives, flavouring agents, artificial sweeteners, dyes, antioxidants, fortifiers, emulsifiers and antifoaming agents.

4.1.1 Antioxidants

Chemicals which are used to prevent oxidation of fats in processed foods such as potato chips, biscuits, breakfast cereals, crackers etc. are called antioxidants. In fact, they are more reactive towards oxygen that the material they are protecting. They also reduce the rate of involvement of free radicals in the ageing process. The two most familiar antioxidants used are butylated hydroxytoluene (BHT) and butylated hydroxyanisole (BHA).

4.1.2 Artificial sweetening agents

Sucrose and fructose are the most widely used natural sweetners. Since they add to our calorie-intake and promote tooth decay, therefore, many people use artificial sweeteners. For example, saccharin, aspartame, alitame, sucralose, cyclamate and L-glucose.

(i) Saccharin is used in form of its sodium or calcium salt which is highly soluble in water. It is not biodegradable and hence does not have calorific value of food. It is primarily used as a sweetening agent by diabetic patients and by those who need to control intake of calories.

(ii) Aspartame : It is the methyl ester of the dipeptide derived from phenylalanine and aspartic acid. Therefore, it is used as a sugar substitute in cold foods and soft drinks.

(iii) Alitame : Although it is similar to aspartame but is more stable than aspartame.

(iv) Sucralose is trichloro derivative of sucrose. It neither provides calories nor causes tooth decay.

(v) Cyclamate : It is N-cyclohexylsulphamate.

(vi) L-Glucose : Like D-sugars, L-sugars are also sweet in taste but do not provide any energy since our body does not have the enzymes for their metabolism.

4.1.3 Preservatives

Chemical substances which are used to protect food against bacteria, yeasts and moulds are called preservatives. The most commonly used preservative is sodium benzoate. Another well known preservative is sodium or potassium metabisulphite which is used in jams, squashes, pickles etc. Its preservative action is due to SO_2 which dissolves in water to form sulphurous acid which inhibits the growth of yeasts, moulds and bacteria. Salts of propionic acid and sorbic acid are also used as preservatives.

4.1.4 Edible colours

Colours are added to thousands of food items to improve their appearance although they do not have any nutritive or food value. These are essentially dyes which may be either synthetic or natural. The synthetic edible colours are azo dyes which are harmful for young children and asthma patients. The most important synthetic dye is tetrazine which is shown to be harmful. Natural edible colours such as annatto, caramel, carotene and saffron are, however, safe. Some inorganic salts have also been used. For example, iron oxide is used to impart red colour and titanium dioxide is used to intensify whiteness.

5. DETERGENTS

5.1 Soaps

Soaps are sodium or potassium salts of higher fatty acids such as lauric acid, ($C_{11}H_{23}COOH$), myristic acid ($C_{13}H_{27}COOH$), palmitic acid ($C_{15}H_{31}COOH$), stearic acid ($C_{17}H_{35}COOH$), oleic acid ($C_{17}H_{33}COOH$), linoleic acid ($C_{17}H_{31}COOH$) and linolenic acid ($C_{17}H_{29}COOH$).

Soaps are 100% biodegradable, i.e., microorganisms present in sewage water completely oxidise them to CO_2. As a result, soaps do not create any water pollution problem. However, they have two disadvantages :

(i) Soaps cannot be used in hard water since calcium and magnesium ions present in hard water produce curdy white precipitates of calcium and magnesium soaps.

(ii) Soaps cannot be used in acidic solutions since acids precipitate the insoluble free fatty acids which adhere to the fabrics and hence prevent the process of dyeing.

5.2 Synthetic detergents

Synthetic detergents, on the other hand, are generally sodium salts of alkyl hydrogen sulphates of long chain alcohols or alkylbenzene sulphonates. Unlike soaps, detergents can be used even in hard water since like sodium or potassium salts, their calcium and magnesium salts are soluble in water. However, unlike soaps, synthetic detergents are not completely biodegradable and hence cause water-pollution. Synthetic detergents are of the following three types :

(a) Anionic detergents : These are so named because a large part of their molecules are anions. These are of two types :

(i) Sodium alkyl sulphate : These are obtained from long chain alcohols by treatment with conc. H_2SO_4 followed by neutralization with NaOH. For example,

$$\underset{\text{n-Lauryl alcohol}}{C_{11}H_{23}CH_2OH} \xrightarrow[(-H_2O)]{\text{Conc. } H_2SO_4} \underset{\text{n-Lauryl hydrogen sulphate}}{C_{11}H_{23}CH_2OSO_2H}$$

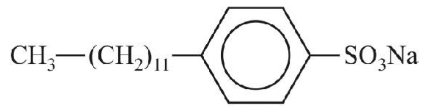
$$\Big\downarrow \text{NaOH } (-H_2O)$$

$$\underset{\text{Sodium lauryl sulphate}}{C_{11}H_{23}CH_2OSO_2Na}$$

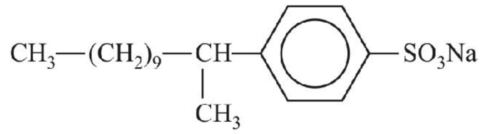
These sodium alkyl sulphates are 100% biodegradable.

(ii) Alkylbenzene sulphonates (ABS) : These are obtained by F.C. alkylation of benzene with long chain alkyl halide or an alkene or an alcohol followed by sulphonation and neutralization with NaOH. The most widely used domestic detergent is the sodium dodecylbenzenesulphonate (SDS).

Another important example is sodium 2-dodecylbenzenesulphonate :



Sod. dodecylbenzenesulphonate



Sod. 2-dodecylbenzenesulphonate

These detergents are also effective in slightly acidic solutions since they form the corresponding alkyl hydrogen sulphates which are soluble materials whereas soaps react with acidic solutions to form insoluble fatty acids.

(b) Cationic detergents : These are quaternary ammonium salts (chlorides, bromides, acetates etc.) containing one or more long chain alkyl groups. Being more expensive than the anionic detergents, they find limited use. Such detergents, however, possess germicidal properties and hence are used quite extensively as germicides, e.g., cetyltrimethylammonium chloride $[CH_3(CH_2)_{15}\overset{\oplus}{N}(CH_3)_3]Cl^{\ominus}$.

(c) Non-ionic detergents : These are esters of high molecular mass obtained by reaction between polyethylene glycol and stearic acid.

These may also be obtained from long chain alcohols by treatment with excess of ethylene oxide in presence of a base.

5.3 Detergent pollution

Detergents having branched hydrocarbon chains cause pollution in rivers and other waterways. The reason being that side chains stop bacteria from attacking and breaking the chains. This results in slow degradation of detergent molecule leading to their accummulation.

ENVIRONMENTAL CHEMISTRY

1. ENVIRONMENTAL POLLUTION

Environmental pollution is an undesirable change in physical, chemical or biological characteristics of our surroundings (air, water or land). It can affect human, animal and plant life as well as materials. Pollution may be natural or man made. It can be classified according to the components of environment being damaged. These are :

(i) Air pollution

(ii) Water pollution

(iii) Soil (land) pollution

2. POLLUTANTS

When the concentration of a substance already present in nature or of a new substance increases to undesirable proportions causing danger to human beings, other animals or vegetation and other materials, the substance is treated as a **pollutant**. The pollutants spoil the environment and are harmful to living organisms and other materials. The common pollutants are :

(i) gases like carbon monoxide, sulphur dioxide, oxides of nitrogen, etc.

(ii) compounds of metals like lead, mercury, zinc, cadmium, arsenic, etc.

(iii) pollen grains, dust

(iv) pesticides and detergents

(v) sewage and

(vi) radioactive substances

3. THRESHOLD LIMIT VALUE (TLV)

This indicates the permissible limit of a pollutant in atmosphere to which a healthy worker is exposed during 8 hours a day or 40 hours a week for life time without any adverse effects. For example, TLV of CO is 50 ppm and that of CO_2 is 5000 ppm. But TLV for a poisonous gas phosgene is only 0.1 ppm.

4. ATMOSPHERE

The gaseous envelope surrounding the earth is known as **atmosphere**. The two major components of dry and clean air in the atmosphere (by volume) are nitrogen (78.09%) and oxygen (20.95%). Argon (0.934%) and carbon dioxide (0.034%) are the minor components of the atmosphere. Air can hold water vapour from 0.1 to 5% by volume and also contains traces of elements such as noble gases (neon, helium, krypton, xenon), hydrogen, methane, sulphur dioxide, ammonia, ozone, etc. The atmosphere surrounding us may be divided into four regions :

(i) **troposphere** (8 to 12 km above earth's surface)

(ii) **stratosphere** (11 to 50 km above earth's surface)

(iii) **mesosphere** (50 to 90 km above earth's surface)

(iv) **thermosphere** (90 to 500 km above earth's surface)

About 80% of the total mass of air and almost all of the water vapours of the atmosphere is found in the inner layer known as **troposphere**.

The chemical and photochemical reactions play a significant role in governing the chemical species present in the atmosphere.

4.1 Air pollutants

There are two types of air pollutants :

1. Primary air pollutants
2. Secondary air pollutants

4.2 Tropospheric pollution

The tropospheric pollution occurs because of the presence of undesirable solid or gaseous particles in air. The pollutants may be broadly classified into two major types :

1. **Gaseous air pollutants :** These include oxides of sulphur, nitrogen and carbon, hydrogen sulphide, hydrocarbons, ozone and other oxidants.

2. **Particulate pollutants :** These are dust, fumes, mist, spray, smoke etc.

4.3 Particulates in air pollution

1. **Particulates :** The small sized solid particles and liquid droplets which range in size from 2×10^{-10} m (0.0002 μm) to 5×10^{-4} m are collectively called as **particulates**. These particles are usually individually not visible to the naked eye. However, small particles often collectively form a haze that restricts the visibility. The common particulates are smoke, mists, fumes, dust etc.

The particulates in the atmosphere may be **viable** or **non-viable**. The **viable particulates** are the small living organisms which are dispersed in the atmosphere. These includes bacteria, moulds, fungi, algae, etc. Some of these viable particulates cause allergic reactions on human beings. Fungi can also cause plant diseases.

Non-viable particulates are formed either by the breakdown of large materials or by the condensation of minute particles and droplets.

The effects of particulate pollutants depend upon the size of the particles. The coarser particles of size more than 5 microns are likely to lodge in the nasal passages whereas the smaller ones are more likely to penetrate into the lungs. The rate of penetration is inversely proportional to the size of the particles. Some of these particles are carcinogens. Continuous inhaling of these small particles for long periods of time irritates the lungs and causes 'scarring' or 'fibrosis' of the lung lining. This type of disease is very common in industrial settings and is known as **"pneumoconiosis"**.

4.4 Smog

Smog is a mixture of smoke, dust particles and small drops of fog. It is a major air pollutant in big cities.

Smog is of two types :

(i) Classical smog, and (ii) Photochemical smog.

1. **Classical smog :** This type of smog is formed by the combination of smoke, dust and fog containing sulphur dioxide from polluted air. This is also called **chemical smog**. Chemically, it is a reducing mixture so it is also called **reducing smog**.

2. **Photochemical smog :** This type of smog is formed by the combination of smoke, dust and fog with an air pollutant in the atmosphere as a result of photochemical reaction.

   The chemistry of formation of photochemical smog centres around nitric oxide (NO). The formation of photochemical smog can be understood by the following steps :

(i) During the early morning before the sun rise, the automobile exhaust emits CO, hydrocarbons and oxides of nitrogen. The NO reacts with oxygen to produce NO_2, a yellowish brown gas.

(ii) As the sun rises, ultra-violet and visible radiations fall on the earth. The ultra-violet rays convert NO_2 back to NO and produce highly reactive atoms of oxygen.

$$NO_2 + \text{u.v. radiation} \longrightarrow NO + \underset{\text{(Reactive oxygen)}}{O}$$

Some of these oxygen atoms combine with O_2 in the air to produce ozone gas.

$$O_2 + O \longrightarrow O_3$$

Ozone reacts with NO to form NO_2 and O_2

$$O_3 + NO \longrightarrow NO_2 + O_2$$

NO_2 again absorbs u.v. radiations and the entire cycle starts again. Both NO_2 and O_3 are strong oxidizing agents and can react with unburnt hydrocarbons (from exhaust of automobiles) to form organic free radicals. The formation of organic free radicals results into a number of chain reactions producing many undesirable compounds (such as formaldehyde, acrolein, organic peroxides, organic hydroperoxides, peroxyacyl nitrates etc.) which constitute photochemical smog. It also includes H_2O_2. The brownish haze of photochemical smog is largely due to brown colour of NO_2.

$$3CH_4 + 2O_3 \longrightarrow \underset{\text{Formaldehyde}}{3HCHO} + 3H_2O$$

Acrolein and peroxyacetylnitrate (PAN) are very noxious substances.

$$\underset{\text{Acrolein}}{CH_2{=}CHCHO} \qquad \underset{\text{Peroxyacetylnitrate (PAN)}}{CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-OONO_2}$$

5. ACID RAIN

The gaseous CO_2 in the atmosphere dissolves in water droplets to form weak acid, carbonic acid.

$$CO_2(g) + H_2O \longrightarrow H_2CO_3(aq)$$

$$H_2CO_3(aq) \rightleftharpoons H^{\oplus}(aq) + HCO_3^{\ominus}(aq)$$

Rain water normally has pH of 5-6 due to the formation of $H^{\oplus}$ ions from the reaction of rain water with carbon dioxide present in the atmosphere. When the pH of the rain water falls below 5-6, it becomes acidic and is called **acid rain**.

However, CO_2 is not the major component of acid rain. The oxides of nitrogen and sulphur undergo many photochemical reactions in atmosphere and form HNO_3 and H_2SO_4 acids. During rains, these acids fall to the earth with rain. This polluted rain is called acid rain.

Acid rain is very damaging. It is toxic to vegetation, human life and aquatic life. Some of its harmful effects are :

1. It causes extensive damage to buildings and sculptural materials of marble, limestone, slate, etc.
2. The acid rain has also caused elimination of life from some fresh water lakes by destroying the living bodies.
3. The rain water also corrodes metals.
4. It damages leaves of trees and plants and retard the growth of forests.

6. DEPLETION OF OZONE LAYER

Ozone is an important constituent of the stratosphere at altitudes between 15 and 25 km. It is formed in the atmosphere by the decomposition of oxygen by ultra-violet radiation from the sun having wavelength shorter than 260 nm.

$$O_2(g) + h\nu \xrightarrow{\text{u.v.}} O(g) + O(g)$$

$$O(g) + O_2(g) \xrightarrow{\text{u.v.}} O_3(g)$$

It is thermodynamically unstable and decomposes into molecular oxygen. It has the important photochemical property of absorbing solar radiation between the wavelength of 200 nm and 300 nm.

$$O_3(g) + h\nu \longrightarrow O(g) + O_2(g)$$

The reactive atomic oxygen formed in the above reaction recombines with molecular oxygen to form ozone. This completes the ozone cycle.

Thus, a dynamic equilibrium exists between the production and decomposition of ozone molecules.

The thick layer of ozone is called **ozone blanket** because it is very effective in absorbing harmful ultra violet rays given out by the sun. Therefore, the ozone layer is also known as **protective shield**.

Recently in 1980, scientists have observed a hole in the ozone blanket covering the upper atmosphere around Antarctica. Recent observations have also shown that the ozone layer diminishes over the south pole in spring during August-September to a greater extent year after year. This depletion of the protective blanket of ozone will cause a damaging effect because harmful ultra-violet rays can reach earth through this hole. The increased level of ultra-violet rays will result in damage to plants, animals, human beings and even matter posing great threat to ecosystem over the globe.

Thus, the depletion of ozone layer is a serious threat to mankind. The depletion of ozone may be due to some natural processes or industrial activities.

The main cause of depletion of ozone layer is its reaction with chlorofluorocarbons. Unlike other chemicals, chlorofluorocarbons are not removed from the atmosphere by usual scavenging processes like photo dissociation, oxidation, rainfall, etc. As a result they move to stratosphere by random diffusion. Then these are destroyed by photolysis and release atomic chlorine. This released Cl atoms cause a catalytic chemical reaction and cause significant depletion of stratospheric ozone layer.

$$Cl_2CF_2 + h\nu \longrightarrow Cl^{\bullet} + ClCF_2^{\bullet}$$

The free radical, Cl• reacts with O_3 through a chain reaction

$$Cl^{\bullet} + O_3 \longrightarrow ClO^{\bullet} + O_2$$

$$ClO^{\bullet} + O \longrightarrow Cl^{\bullet} + O_2$$

These chlorine atoms are free to react with more ozone. As a result, many O_3 molecules can be destroyed for each chlorine atom produced. It has been shown that over one thousand ozone molecules can be destroyed by one Cl.

The net result of these reactions is destruction of several molecules of O_3 for each Cl atom produced.

7. GREENHOUSE EFFECT

The heating of the earth due to trapped radiation is called **greenhouse effect**.

The gases which can trap infra-red radiation given by the sun to produce greenhouse effect leading to heating up the environment are called **greenhouse gases**. The principal gases of the atmosphere N_2, O_2 and Ar do not absorb infra-red light. The common atmospheric gases that have predominantly produced green house warming are CO_2, water vapours (H_2O) and ozone (O_3).

8. WATER POLLUTION

Water pollution may be defined as any change in its physical, chemical or biological properties or contamination with foreign materials that can adversely affect human beings or reduce its utility for the intended use.

The major water pollutants and their common sources are given below :

Pollutant	Sources
Micro organisms	Domestic sewage
Organic wastes	Domestic sewage, animal manures, food processing wastes, decaying plants and animals, paper discards, rags and other bio degradable wastes.
Plant nutrients	Chemical fertilizers.
Toxic heavy metals	Industries and chemical factories
Sediments	Insoluble particles of soil and rock and inorganic and organic compounds which wash into water from mining, agricultural activities, construction activities and soil erosion.
Pesticides	Chemicals used for killing insects, fungi and weeds.
Radioactive wastes	Mining of uranium containing minerals and accidental release from nuclear plants.
Heat	Water used by industrial plants for cooling which is discharged as hot water.

9. GREEN CHEMISTRY

It means zero discharge of toxic substances into the environment guaranted by the fact that they are never produced. It may be defined as the strategy to design chemical processes and products that reduces or eliminates the use and generation of hazardous substances. So **Green chemistry prevents problems before they occur by designing new approaches.**

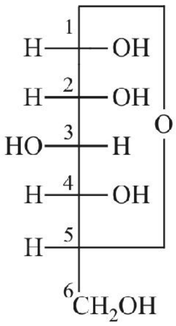
Note...

1. Biological Oxygen Demand (BOD) : Amount of dissolved oxygen needed by aerobic bacteria to break down the organic matter.
2. Chemical Oxygen Demand (COD) : Amount of oxygen which reacts with chemicals other than organic wastes.
3. BOD measures oxygen consumed by living organisms assimilating organic matter present in waste.
4. COD measures biological oxidisable as well as biological inert organic matter such as cellulose.
5. If BOD and COD values are large, water is heavily polluted.

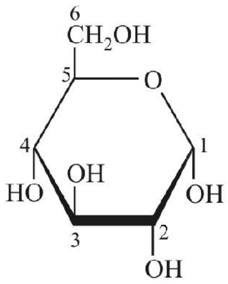
APPENDIX

Chair conformations of various carbohydrates

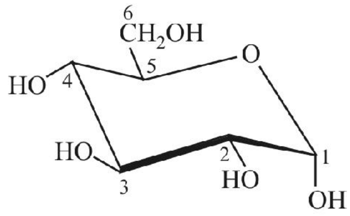
1. α-D-glucose



Fischer projection

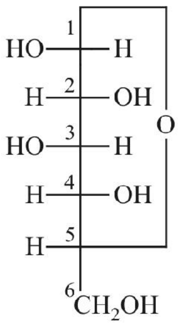


Haworth Projection
α-D-Glucose

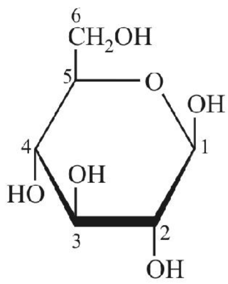


Chair conformation

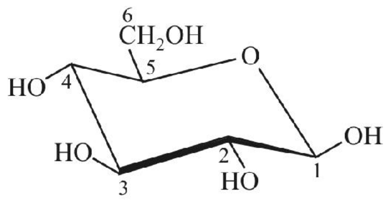
2. β-D-glucose



Fischer projection



Haworth Projection
β-D-Glucose



Chair conformation

EXERCISE–1

(Basic Level Objective Questions)

BIOMOLECULES

1. A disaccharide on hydrolysis gives
(a) Two molecules of the same monosaccharide
(b) One molecule each of two different monosaccharides
(c) Three molecules of the same monosaccharide
(d) Two molecules of the same or one molecule each of the two different monosaccharides.

2. Glucose is a/an
(a) Aldohexose (b) Aldopentose
(c) Aldotetrose (d) Ketohexose

3. Sucrose reacts with acetic anhydride to form
(a) Penta-acetate (b) Hexa-acetate
(c) Tetra-acetate (d) Octa-acetate

4. All monosaccharides containing five or six carbon atoms have
(a) Open chain structures
(b) Pyranose structures
(c) Furanose structures
(d) May have pyranose or furanose structures

5. Which of the following is a reducing sugar ?
(a) Glucose (b) Fructose
(c) Maltose (d) All of these

6. Which of the following reduces Tollen's reagent ?
(a) Glucose (b) Fructose
(c) Lactose (d) All

7. Which of the following is a non-reducing sugar ?
(a) Sucrose (b) Maltose
(c) Lactose (d) Ribose

8. Glucose reduces
(a) Tollen's reagent (b) Fehling's solution
(c) Benedict's solution (d) All

9. Which of the following form/s osazone with phenylhydrazine ?
(a) Glucose (b) Fructose
(c) Maltose (d) All the three above

10. Glucose and fructose are
(a) Optical isomers (b) Tautomers
(c) Functional isomers (d) Chain isomers

11. Glucose and mannose are
(a) Optical isomers (b) Anomers
(c) Epimers (d) Chain isomers

12. Acid or enzymatic hydrolysis of sucrose to give an equimolar mixture of glucose and fructose is called
(a) Esterification (b) Inversion
(c) Saponification (d) Insertion

13. Lactose is a disaccharide of
(a) Glucose only (b) Glucose and fructose
(c) Glucose and galactose
(d) Galactose only

14. Proteins are condensation polymers of
(a) α-Amino acids (b) β-Amino acids
(c) α-Hydroxy acids (d) β-Hydroxy acids

15. Proteins are
(a) Polyamides (b) Polyesters
(c) Polyhydric alcohols (d) Polycarboxylic acids

16. The peptide bond is
(a) $-CONH_2$ (b) $-CONH-$
(c) $-COONH_4$ (d) $-N=C=O$

17. Which of the following α-amino acids does not contain a chiral carbon ?
(a) Glycine (b) Alanine
(c) Phenylalanine (d) Valine

18. In aqueous solutions, amino acids mostly exist as
(a) $NH_2—CHR—COOH$
(b) $NH_2—CHR—COO^{\ominus}$
(c) $N_3\overset{\oplus}{N}—CHR—COOH$
(d) $H_3\overset{\oplus}{N}—CHR—COO^{\ominus}$

19. The sequence in which the α-amino acids are linked to one another in a protein molecule is called its
(a) Primary structure (b) Secondary structure
(c) Tertiary structure (d) Quaternary structure

20. Mark the wrong statement about denaturation of proteins
(a) The primary structure of the protein doesn't change
(b) Globular proteins are converted into fibrous proteins
(c) Fibrous proteins are converted into globular proteins
(d) The biological activity of the protein is destroyed

21. The sugar present in DNA is
(a) Glucose (b) Deoxyribose
(c) Ribose (d) Fructose

22. The pentose sugar in DNA and RNA has the
(a) Open chain structure (b) Pyranose structure
(c) Furanose structure (d) None of the above

23. Which of the following sets of bases is present both in DNA and RNA ?
(a) Adenine, uracil, thymine
(b) Adenine, guanine, cytosine
(c) Adenine, guanine, uracil
(d) Adenine, guanine, thymine

24. Consider the double helix structure of DNA. Are the base pairs
(a) Part of the back bone structure
(b) Inside the helix
(c) Outside the helix
(d) None of these

25. Which of the following statements about DNA is not correct ?
(a) It has a double helix structure.
(b) It undergoes replication.
(c) The two strands in a DNA molecule are exactly similar.
(d) It contains the pentose sugar, 2-deoxyribose.

26. Which of the following statements about RNA is not correct ?
(a) It has a single strand.
(b) It does not undergo replication.
(c) It does not contain any pyridimine base.
(d) It controls the synthesis of proteins.

27. If the sequence of bases in DNA is TGAACCCTT then the sequence of bases in m-RNA is
(a) ACUUGGGAA (b) TCUUGGGTT
(c) ACUUCCCAA (d) None of the above

28. The chemical messengers produced in the ductless glands are
(a) Vitamins (b) Lipids
(c) Hormones (d) Antibodies

29. Which of the following hormones contains iodine ?
(a) Insulin (b) Thyroxine
(c) Adrenaline (d) Testosterone

30. Insulin is a
(a) Steroid hormone (b) Peptide hormone
(c) Amine hormone (d) None of the above

31. The bond that determines the secondary structure of a protein is
(a) Co-ordinate bond (b) Covalent bond
(c) Hydrogen bond (d) Ionic bond

32. Which of the following contains nitrogen ?
(a) Fats (b) Proteins
(c) Carbohydrates (d) None

33. The reagent which forms crystalline osazone derivative when treated with glucose is
(a) Fehling solution (b) Phenylhydrazine
(c) Benedict solution (d) Hydroxylamine

34. In nucleic acids, the sequence is
(a) Phosphate–Base–sugar
(b) Sugar–Base–phosphate
(c) Base–sugar–phosphate
(d) Base–phosphate–sugar

35. The pH of blood is around
(a) 8.0 (b) 6.0
(c) 5.4 (d) 7.4

36. α-D-Glucose and β-D-glucose differ from each other due to difference in one carbon with respect to its ?
(a) Size of hemiacetal ring (b) Number of OH groups
(c) Configuration (d) Conformation

37. In nucleic acids, the nucleotides are linked to one another through
(a) Hydrogen bond (b) Peptide bond
(c) Glycosidic linkage (d) Phosphate groups

38. If one strand of DNA has the sequence ATGCTTGA, the sequence in the complimentary strand would be
(a) TACGAACT (b) TCCGAACT
(c) TACGTACT (d) TACGTAGT

39. When amylases catalyse the hydrolysis of starch, the final product obtained is chiefly
(a) Cellubiose (b) Glucose
(c) Maltose (d) Sucrose

40. Man dies in an atmosphere of carbon monoxide because

(a) It increases the blood pressure.

(b) It combines with oxygen present in the body to form carbon dioxide.

(c) It combines with the haemoglobin of the blood making it incapable of absorbing oxygen.

(d) It reduces the organic matter.

41. The acid showing salt-like character in aqueous solution is

(a) Acetic acid (b) Benzoic acid

(c) Formic acid (d) α-Aminoacetic acid

42. In polysaccharides, the linkage connecting monosaccharides is called

(a) glycoside linkage (b) nucleoside linkage

(c) glycogen linkage (d) peptide linkage

43. A nucleotide consists of

(a) carbon sugar (b) nitrogen containing base

(c) phosphoric acid (d) all of these

44. In an amino acid, the carboxyl group ionises at $pKa_1 = 2.34$ and ammonium ion at $pKa_2 = 9.60$. The isoelectric point of the amino acid is at pH

(a) 5.97 (b) 2.34

(c) 9.60 (d) 6.97

45. In fructose, the possible optical isomers are

(a) 12 (b) 8

(c) 16 (d) 4

46. Which is not a true statement ?

(a) α-carbon of α-amino acid is asymmetric

(b) all proteins are found in L-form

(c) human body can synthesize all proteins they need

(d) at pH = 7 both amino and carboxylic groups exist in the ionised form

47. The number of amino acids found in proteins that a human body can synthesize is

(a) 20 (b) 10

(c) 5 (d) 14

48. Oxidation of glucose with Ag_2O gives

(a) D-Gluconic acid (b) L-Glucaric acid

(c) L-Gluconic acid (d) L-Glucaric acid

49. In lactose, the reducing part is

(a) Galactose (b) Glucose

(c) Fructose (d) Mannose

50. In glycine, the basic group is

(a) $-NH_2$ (b) $-NH_3^{\oplus}$

(c) $-COOH$ (d) $-COO^{\ominus}$

51. The relation of the isoelectric point for an amino acid, to solubility is

(a) the two are not related

(b) an amino acid is least soluble at its isoelectric point

(c) an amino acid has the maximum solubility at the isoelectric point

(d) solubilities of only some amino acids depend on it.

52. Which of the following does not exist as a zwitterion ?

(a) Glycine (b) Alanine

(c) Sulphanilic acid (d) p-Aminobenzoic acid

53. A certain compound gives negative test with ninhydrin and positive test with Benedict's solution. The compound is

(a) A protein (b) A monosacchride

(c) A lipid (d) An amino acid

54. The number of tripeptides formed by three different amino acids are

(a) Three (b) Four

(c) Five (d) Six

55. The dipeptide glycylalanine contains

(a) glycine as C-terminal residue

(b) glycine as N-terminal residue

(c) alanine as N-terminal residue

(d) either (a) or (b)

56. Which statement correctly completes the statement ?

Except for glycine, which is achiral, all the amino acids present in proteins ....

(a) are chiral, but racemic

(b) have the L configuration at their α carbon

(c) have the R configuration at their α carbon

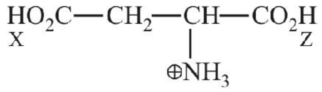
(d) have the S configuration at their α carbon

57. Assume that a particular amino acid has an isoelectric point of 6.0. In a solution at pH 1.0, which of the following species will predominate ?

(a) $H_3\overset{\oplus}{N}\overset{\overset{R}{|}}{C}HCO_2H$ (b) $H_2N\overset{\overset{R}{|}}{C}HCO_2H$

(c) $H_3\overset{\oplus}{N}\overset{\overset{R}{|}}{C}HCO_2^{\ominus}$ (d) $H_2N\overset{\overset{R}{|}}{C}HCO_2^{\ominus}$

58. The pK_a values for the three ionizable groups X, Y and Z of glutamic acid are 4.3, 9.7 and 2.2 respectively

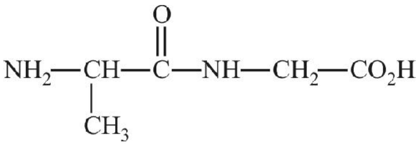


The isoelectric point for the amino acid is

(a) 7.00 (b) 3.25

(c) 4.95 (d) 5.95

59.



Identify the amino acid obtained by hydrolysis of the above compound.

(a) Glycine (b) Alanine

(c) Both (a) and (b) (d) None of these

60. At Iso-electric point

(a) conc. of cation is equal to conc. of anion

(b) Net charge is zero.

(c) Maximum conc. of di-polar ion (Zwitter ion) will be present

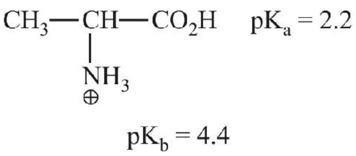
(d) All of the above

61. Which of following amino acid has lowest iso-electric point ?

(a) Glycine (b) Alanine

(c) Aspartic acid (d) Lysine

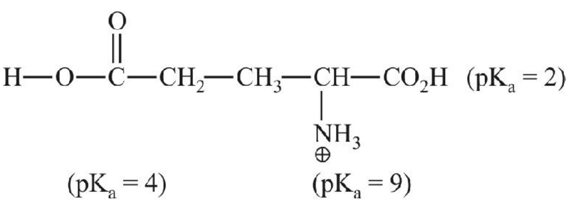
62. Find iso-electric point of given amino acid



(a) 3.3 (b) 5.9

(c) 9.6 (d) 11.8

63. Find iso-electric point of the given amino acid



(a) 5.5 (b) 6.5

(c) 3 (d) 5

64. $H-C\equiv C-H \xrightarrow[H_2SO_4]{HgSO_4} (A) \xrightarrow[(2)\ H_3O^{\oplus}]{(1)\ NH_3 + HCN} (B)$;

Product (B) of above reaction is

(a) Glycine (b) Alanine

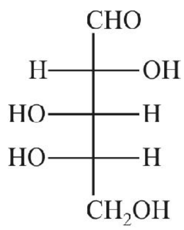
(c) Valine (d) Leucine

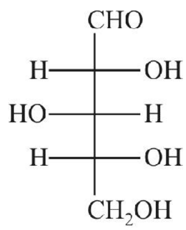
65. Which amino acid does not contain chiral centre ?

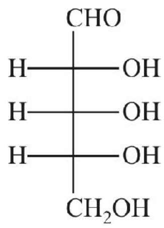
(a) Valine (b) Leucine

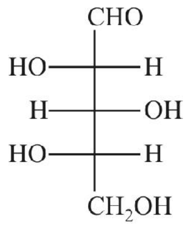
(c) Glycine (d) Iso-leucine

66. Which L-sugar on oxidation gives an optically active dibasic acid (2 COOH groups) ?

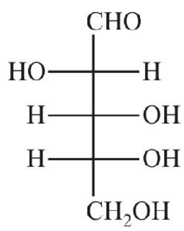
(a) 

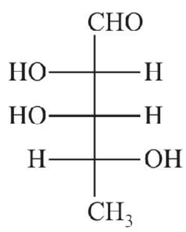
(b) 

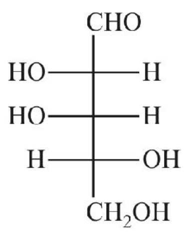
(c) 

(d) 

67. Among the three compounds shown below, two yield the same product on reaction with warm HNO_3. The exception is

(a) 

(b) 

(c) 

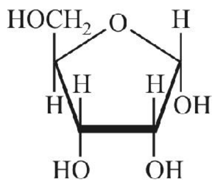
(d) None of these

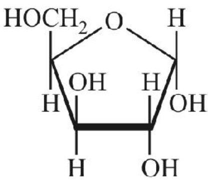
68. The optical rotation of the α-form of a pyranose is +150.7°, that of the β-form is +52.8°. In solution an equilibrium mixture of these anomers has an optical rotation of +80.2°. The percentage of the α form in equilibrium mixture is

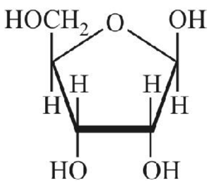
(a) 28% (b) 32%

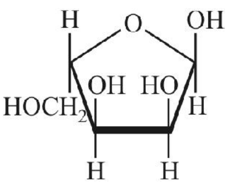
(c) 68% (d) 72%

69. Which of the following represents the anomer of the compound shown ?



(a) 

(b) 

(c) 

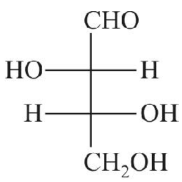
(d) None of these

70. For the complex conversion of D-glucose into the corresponding osazone, the minimum number of equivalents of phenyl hydrazine required is

(a) two (b) three

(c) four (d) five

71. The configurations of the chirality centres in D-threose (shown) are



(a) 2R, 3R (b) 2R, 3S

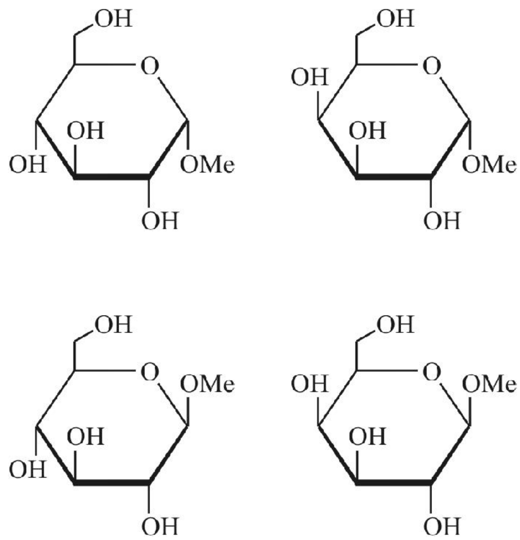
(c) 2S, 3R (d) 2S, 3S

72. Rapid interconversion of α-D-glucose and β-D-glucose in solution is known as

(a) racemization (b) asymmetric induction

(c) fluxional isomerization (d) mutarotation

73. Identify the correct set of stereochemical relationships amongst the following monosaccharides I-IV.



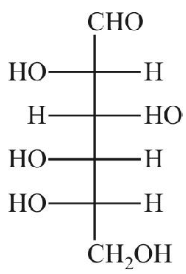
(a) I and II are anomers ; III and IV are epimers

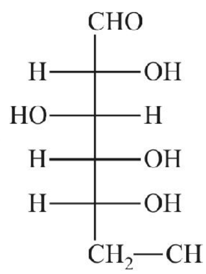
(b) I and III are epimers ; II and IV are anomers

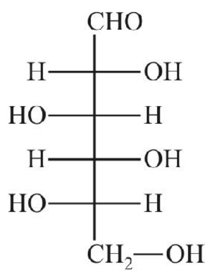
(c) I and II are epimers ; III and IV are anomers

(d) I and III are anomers ; I and II are epimers

74. What is the structure of L-Glucose ?

(a) 

(b) 

(c) 

(d) None of these

75. Stereoisomers of aldoheptose is (a) and stereoisomers of ketoheptose is (b). Ratio of a/b is

(a) $\frac{1}{2}$ (b) $\frac{2}{1}$

(c) $\frac{4}{1}$ (d) $\frac{1}{4}$

76. D-Glucose $\xrightarrow[\Delta]{HNO_3}$ (A), Product (A) is

(a) D-Gluconic acid (b) D-Glucitol

(c) D-Fructose (d) D-Glucaric acid

77. D-glucose & D-fructose can be differentiated by

(a) Fehling sol

(b) Tollens reagent

(c) Benedict test

(d) Br_2/H_2O

78. D-Glucose exists in x different forms. The value of x (stereoisomer) is

(a) 2 (b) 3

(c) 4 (d) 5

POLYMERS

79. Which of the following is a natural polymer ?

(a) Protein (b) Polythene

(c) Buna-S (d) Bakelite

80. The monomers used in the manufacture of nylon-6,6 are

(a) sebacic acid and hexamethylene diamine

(b) adipic acid and butadiene

(c) sebacic acid and butadiene

(d) adipic acid and hexamethylene diamine

81. PVC polymer can be prepared by which of the monomers ?

(a) $CH_3CH{=}CH_2$ (b) $C_6H_5CH{=}CH_2$

(c) $CH_2{=}CHCl$ (d) $CH_2{=}CH_2$

82. Which of the following sets contains only addition polymers ?

(a) Polyethylene, polypropylene, terylene

(b) Polyethylene, PVC, acrilon

(c) Buna-S, nylon, polybutadiene

(d) Bakelite, PVC, polyethylene

83. Nylon is

(a) polyester fibre

(b) polyamide fibre

(c) polythene derivative

(d) polyethylene methyl acrylate fibre

84. Which one of the following statements is wrong ?

(a) PVC stands for poly vinyl chloride

(b) PTFE stands for teflon

(c) PMMA stands for polymethyl metha acrylate

(d) Buna-S stands for natural rubber

85. Which of the following fibres are made of polyamides ?

(a) Dacron (b) Orlon

(c) Nylon (d) Rayon

86. Which of the following is not an example of natural polymer ?

(a) Wool (b) Silk

(c) Leather (d) Nylon

87. Natural rubber is a polymer of

(a) neoprene (b) isoprene

(c) chloroprene (d) butadiene

88. Heating of rubber with sulphur is known as

(a) galvanisation (b) bessemerisation

(c) vulcanisation (d) sulphonation

89. Buna-S is a polymer of

(a) butadiene (b) butadiene and styrene

(c) styrene (d) butadiene and chloroprene

90. Polythene is a resin obtained by polymerization of

(a) butadiene (b) ethylene

(c) isoprene (d) styrene

91. Bakelite is obtained from phenol by reacting with

(a) formaldehyde (b) acetaldehyde

(c) chlorobenzene (d) acetal

92. Dacron is an example of

(a) polyamide (b) polypropylene

(c) polyurethane (d) polyester

93. Orlon has a unit

(a) isoprene (b) acrolein

(c) glycol (d) vinylcyanide

94. The monomer of teflon is

(a) $CHF{=}CH_2$ (b) $CHF{=}CHCl$

(c) $CHCl{=}CHCl$ (d) $CF_2{=}CF_2$

95. The fibre obtained by the condensation of hexamethylene diamine and adipic acid is

(a) dacron (b) nylon-6,6

(c) rayon (d) teflon

96. Which one of the following is a thermosetting polymer ?

(a) Nylon-6 (b) Nylon-6,6

(c) Bakelite (d) SBR

97. The widely used plastic PVC is a polymerization product of

(a) $CH_2{=}CH_2$ (b) $CH_2{=}CCl_2$

(c) $CHCl{=}CHCl$ (d) $CH_2{=}CHCl$

98. Polymer obtained by condensation polymerization is

(a) polythene (b) teflon

(c) PVC (d) phenol-formaldehyde resin

99. Which of the following is not a biopolymer ?

(a) Starch (b) Rubber

(c) Proteins (d) Nucleic acids

100. Which of the following is chain growth polymer ?

(a) Polypropylene (b) Glyptal

(c) Nylon-6,6 (d) Nylon-6

101. An example of natural biopolymer is

(a) nylon (b) rubber

(c) teflon (d) DNA

102. The product of addition polymerization reaction is

(a) nylon (b) teflon

(c) polythene (d) terylene

103. Which of the following is an example of condensation polymer ?

(a) Buna-S rubber (b) Bakelite

(c) Nylon-6,6 (d) All of these

104. Terylene is a condensation polymer of ethylene glycol and

(a) phthalic acid (b) terephthalic acid

(c) benzoic acid (d) salicyclic acid

105. Neoprene is a polymer of

(a) isoprene (b) butadiene

(c) styrene (d) chloroprene

106. A polyamide synthetic polymer prepared by prolonged heating of caprolactum, is

(a) nylon-6,6 (b) nylon-6

(c) nylon-6,10 (d) glyptal

107. A raw material used in making nylon is

(a) ethylene (b) butadiene

(c) adipic acid (d) isoprene

108. Nylon threads are made of

(a) polyvinyl polymer

(b) polyester polymer

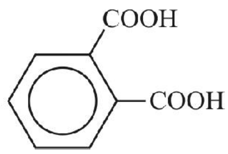
(c) polyamide polymer (d) polyethylene polymers

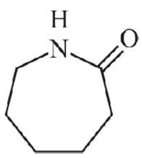
109. Which one of the following is a chain growth polymer ?

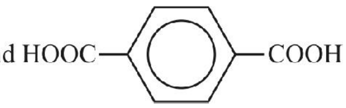
(a) starch (b) nucleic acid

(c) polystyrene (d) protein

110. The monomer of dacron is/are

(a) $HOCH_2{-}CH_2OH$ and 

(b) 

(c) $HOCH_2{-}CH_2OH$ and 

(d) $F_2C{=}CF_2$

111. Which of the following polymer has an ester linkage ?

(a) PVC (b) nylon-66

(c) SBR (d) terylene

112. Nylon is not a

(a) condensation polymer (b) polyamide

(c) copolymer (d) homopolymer

113. Which of the following is used in vulcanization of rubber ?

(a) SF_6 (b) CF_4

(c) C_2F_4 (d) C_2F_2

114. A polymer containing nitrogen is

(a) bakelite (b) dacron

(c) rubber (d) nylon-6,6

115. In elastomer, intermolecular forces are

(a) nil (b) weak

(c) strong (d) very strong

116. Which of the following is a bio-degradable polymer ?
(a) PVC (b) Nylon-6
(c) Cellulose (d) Polythene

117. Which of the following is a polyamide ?
(a) Teflon (b) Nylon-6,6
(c) Terylene (d) Bakelite

118. Buna-S is a polymer of
(a) butadiene only (b) butadiene and nitryl
(c) styrene only (d) butadiene and styrene

119. Condensation product of caprolactum is
(a) nylon-6 (b) nylon-6,6
(c) nylon-60 (d) nylon-6,10

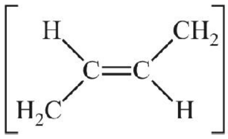
120. Which of the following is a cross-linked polymer ?
(a) Teflon (b) Orlon
(c) Nylon (d) Bakelite

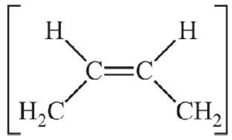
121. $[NH(CH_2)_6NHCO(CH_2)_4CO]_n$ is a
(a) thermosetting polymer (b) homopolymer
(c) copolymer (d) addition polymer

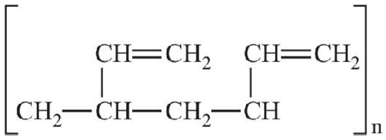
122. Which is not a polymer ?
(a) Sucrose (b) Enzyme
(c) Starch (d) Teflon

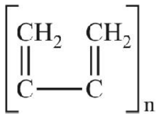
123. Which of the following is true for polypropylene ?
(a) Propylene, condensation polymer
(b) Propylene, addition polymer
(c) Propylene, anionic polymer
(d) Propylene, cationic polymer

124. Mark out the most unlike form of polymerization of
$H_2C{=}CH{-}CH{=}CH_2$

(a) 

(b) 

(c) 

(d) 

125. Which of the following are copolymers ?
(a) Bakelite (b) Melamine
(c) Buna-S (d) Nylon-6,6

CHEMISTRY IN EVERYDAY LIFE

126. Which of the following is an antibiotic ?
(a) Terramycin (b) Aspirin
(c) Paracetamol (d) Chloroquine

127. Which of the following is used as antipyretic ?
(a) Paracetamol (b) Chloroquine
(c) Chloramphenicol (d) LSD

128. Which of the following groups would you introduce into a dye to make it water soluble ?
(a) $-NO_2$ (b) $-Cl$
(c) $-SO_3H$ (d) $-OH$

129. Which one is an example of vat dye ?
(a) Congo red (b) Alizarin
(c) Malachite green (d) Indigo

130. With which of the following cations, alizarin will impart a violet colour on the fabrics ?
(a) Fe^{3+} (b) Cr^{3+}
(c) Ba^{2+} (d) Al^{3+}

131. Aspirin is a/an
(a) narcotic (b) antipyretic
(c) tranquillizer (d) anaesthetic

132. The antibiotic used for the treatment of typhoid is
(a) penicillin (b) chloramphenicol
(c) terramycin (d) sulphadiazine

133. Which one of the following is a chromophore group ?
(a) $-N{=}N-$ (b) $-OH$
(c) $-SO_3H$ (d) $-NH_2$

134. A substance which can act both as an antiseptic and disinfectant is
(a) aspirin (b) phenol
(c) analgin (d) sodium pentothal

135. The substances which relieve anxiety, reduce mental tension and induce sleep are called
(a) analgesics (b) antipyretics
(c) tranquillizer (d) anaesthetics

136. Dettol consists of
(a) cresol + ethanol (b) xylenol + terpeneol
(c) chloroxylenol + terpeneol
(d) none of these

137. Sulpha drugs are used for
(a) removing bacteria (b) precipitating bacteria
(c) stopping the growth of bacteria
(d) decreasing the size of bacteria

138. Morphine is
(a) antiseptic (b) analgesic
(c) antibiotic (d) anaesthetic

139. Which is not antipyretic ?
(a) novalgin (b) aspirin
(c) paracetamol (d) irgapyrin

140. Gammexane is
(a) chlorobenzene (b) DDT
(c) benzene hexa chloride (d) none of these

141. Which of the following acts as chromophore group ?
(a) carbonyl or thio carbonyl
(b) azo or azoxy
(c) nitro or nitroso (d) all of these

142. Indigo belongs to
(a) direct dyes (b) vat dyes
(c) ingrain dyes (d) mordant dyes

143. The compound used to fix a dye to the fabric is known as
(a) mordant (b) bleaching agent
(c) azeotrope (d) leuco base

144. The number of chromophores in picric acid is
(a) 1 (b) 2
(c) 3 (d) 4

145. Which of the following is a plasticizer ?
(a) coconut oil (b) mustard oil
(c) castor oil (d) pine oil

146. Match List I with List II and select the correct answer using the codes given below the list.

List I	List II
I. Iodoform	A. Anaesthetic
II. Methyl salicylate	B. Antiseptic
III. Diethyl ether	C. Insecticide
IV. Hexachlorocyclohexane	D. Detergent
	E. Pain balm

Codes :
(a) I-B, II-E, III-C, IV-D (b) I-D, II-B, III-A, IV-C
(c) I-B, II-E, III-A, IV-C (d) I-C, II-A, III-D, IV-B

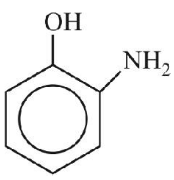
147. Which of the following represents a biliquid propellant ?
(a) nitro glycerine + nitro cellulose
(b) N_2O_4 + unsymmetrical dimethyl hydrazine
(c) N_2O_4 + acrylic rubber
(d) polybutadiene + ammonium perchlorate

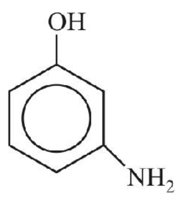
148. The dyes which are applied to the fabric in the colourless reduced state and then oxidised to coloured state are called
(a) vat dyes (b) disperse dyes
(c) triphenyl methane dyes
(d) azo dyes

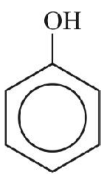
149. Which of the following is a mordant dye ?
(a) aniline black (b) congo red
(c) alizarin (d) indigo

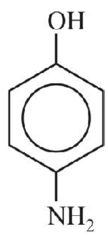
150. Which one of the following is not a broad spectrum antibiotic ?
(a) Tetracycline (b) Chloromycetin
(c) Penicillin (d) None of these

151. Which of the following gives paracetamol on acetylation ?

(a) 

(b) 

(c) 

(d) 

152. The hybrid rocket propellant consists of
(a) acrylic rubber and liquid nitrogen tetroxide
(b) polyurethane and ammonium perchlorate
(c) nitroglycerine and nitrocellulose
(d) liquid hydrogen and liquid oxygen
(e) hydrogen peroxide

153. The basic dye among the following is
(a) Orange-1
(b) Aniline yellow
(c) Anthraquinone
(d) Phenolphthalein

154. An insoluble dye is reduced to a soluble colourless leuco form by an alkaline reducing agent. The fibre is soaked in the dye solution and then exposed to air to develop the colour. The dye is
(a) mordant dye (b) vat dye
(c) azo dye (d) direct dye
(e) disperse dye

155. Tincture of iodine is
(a) aqueous solution of I_2
(b) alcoholic solution of I_2
(c) solution of I_2 in aqueous KI
(d) aqueous solution of KI

ENVIRONMENTAL CHEMISTRY

156. The substance which is a primary pollutant ?
(a) H_2SO_4 (b) CO
(c) PAN (d) Aldehydes

157. The greatest affinity for haemoglobin is shown by which of the following
(a) NO (b) CO
(c) O_2 (d) CO_2

158. Which of the following is not involved in the formation of photochemical smog ?
(a) Hydrocarbon (b) NO
(c) SO_2 (d) O_3

159. Formation of London smog takes place in
(a) Winter during day time
(b) Summer during day time
(c) Summer during morning time
(d) Winter during morning time

160. Which of the following statement is false ?
(a) London smog is oxidising in nature
(b) Photochemical smog causes irritation in eyes
(c) London smog is a mixture of smoke and fog
(d) Photochemical smog results in the formation of PAN

161. The viable particulate among the following is
(a) Fumes (b) Algae
(c) Smoke (d) Mist

162. The non-viable particulate among the following is
(a) Dust (b) Bacteria
(c) Moulds (d) Fungi

163. Which forms the part of hazy fumes of photochemical smog
(a) SO_2 (b) Nitrogen dioxide
(c) PAN formation (d) Aldehydes

164. In which of the following the ozone layer is present
(a) Exosphere (b) Troposphere
(c) Stratosphere (d) Mesosphere

165. Depletion of ozone layer causes
(a) Breast cancer (b) Blood cancer
(c) Lung cancer (d) Skin cancer

166. The gas responsible for ozone depletion
(a) NO and freons (b) SO_2
(c) CO_2 (d) CO

167. The reaction caused by U.V. radiations from sun produces
(a) Ozone (b) Carbon monoxide
(c) Sulphur dioxide (d) Fluorides

168. Which of the following chemical, harmful to ozone, is released by chlorofluoro carbon
(a) Sulphur dioxide
(b) Fluorine
(c) Chlorine
(d) Nitrogen dioxide

169. Ozone hole refers to
(a) Increase in concentration of ozone
(b) Hole in ozone layer
(c) Reduction in thickness of ozone layer in troposphere
(d) Reduction in thickness of ozone layer in stratsophere

170. Ozone layer of stratosphere requires protection from indiscriminate use of
(a) Baloons (b) Pesticides
(c) Atomic explosions (d) Aerosols and high flying jets

171. Which of the following statements about polar stratosphere clouds (PSCs) is not correct ?
(a) PSCs do not react with chlorine nitrate and HCl.
(b) Type I clouds are formed at about –77°C and contain solid HNO_3. $3H_2O$.
(c) Type II clouds are formed at about –85°C and contain some ice.
(d) A tight whirlpool of wind called Polar Vortex is formed which surrounds Antarctica.

172. Phosphate fertilizers when added to water leads to
(a) Increased growth of decomposers
(b) Reduced algal growth
(c) Increased algal growth
(d) Nutrient enrichment (eutrophication)

173. The type of pollution caused by spraying of DDT
(a) Air and soil (b) Air and water
(c) Air (d) Air, water and soil

174. Which is true about DDT
(a) Greenhouse gas (b) A fertilizer
(c) Biodegradable pollutant
(d) Non-biodegradable pollutant

175. Minamata disease is due to pollution of
(a) Arsenic into the atmosphere
(b) Organic waste into drinking water
(c) Oil spill in water
(d) Industrial waste mercury into fishing water

176. BOD is connected with
(a) Microbes and organic matter
(b) Organic matter
(c) Microbes (d) None of these

177. Eutrophication causes reduction in
(a) Dissolved oxygen (b) Nutrients
(c) Dissolved salts (d) All the above

178. Phosphate pollution is caused by
(a) Sewage and agricultural fertilizers
(b) Weathering of phosphate rocks only
(c) Agricultural fertilizers only
(d) Phosphate rocks and sewage

179. Water is often treated with chlorine to
(a) Remove hardness (b) Increase oxygen content
(c) Kill germs (d) Remove suspended particles

180. Ozone is an important constituent of stratosphere because it
(a) Destroys bacteria which are harmful to human life
(b) Prevents the formation of smog over large cities
(c) Removes poisonous gases of the atmosphere by reacting with them
(d) Absorbs ultraviolet radiation which is harmful to human life

181. Presence of which fuel gas in the exhaust fumes shows incomplete combustion of fuel
(a) Sulphur dioxide
(b) Carbon monoxide and water vapour
(c) Carbon monoxide (d) Nitrogen dioxide

182. Which one of the following statements about ozone and ozone layer is true
(a) Ozone layer is beneficial to us because ozone cuts out the ultraviolet radiation of the sun.
(b) The conversion of ozone to oxygen is an endothermic reaction.
(c) Ozone has a triatomic linear molecule.
(d) None of these

183. The ozone layer is present in
(a) Stratosphere (b) Troposphere
(c) Thermosphere (d) Mesosphere

184. The green house effect is caused by
(a) CO_2 (b) NO_2
(c) NO (d) CO

185. Formation of ozone in the upper atmosphere from oxygen takes place by the action of
(a) Nitrogen oxides (b) Ultraviolet rays
(c) Cosmic rays (d) Free radicals

186. When rain is accompanied by a thunderstorm, the collected rain water will have a pH value
(a) Slightly lower than that of rain water without thunderstorm
(b) Slightly higher than that when the thunderstorm is not there
(c) Uninfluenced by occurrence of thunderstorm
(d) Which depends upon the amount of dust in air

QUALITATIVE AND QUANTITATIVE ANALYSIS

187. In the Lassaigne's test for the detection of nitrogen in an organic compound, the blue or green colour is due to the formation of

(a) $K_3[Fe(CN)_6]$ (b) $K_4[Fe(CN)_6]$

(c) $Fe(CN)_6$ (d) $Fe_4[Fe(CN)_6]_3$

188. In the Lassaigne's test, the blood red colouration is due to the formation of

(a) $Fe(CNS)_2$ (b) NaCNS

(c) NH_4CNS (d) $Fe(CNS)_3$

189. A violet colour with sodium nitroprusside in the test of sulphur in an organic compound is due to the formation of

(a) $Na_4[Fe(CN)_5NO]$ (b) $Na_4[Fe(CN)_5NOS]$

(c) $Na_3[Fe(CN)_5NS]$ (d) $Fe_4[Fe(CN)_6]_3$

190. For detection of sulphur in an organic compound, the Lassaigne's extract is acidified with dilute acetic acid and lead acetate is then added to it. The black ppt. is obtained which is due to the formation of

(a) PbS (b) Na_2S

(c) $PbSO_4$ (d) None ofthese

191. Which of the following methods is used to estimate nitrogen in the laboratory ?

(a) Kjeldahl's method (b) Duma's method

(c) Liebig's method (d) Carius method

192. Kjeldahl's method cannot be used for the estimation of nitrogen in

(a) Pyridine (b) Nitro compounds

(c) Azo compounds (d) All the above

193. In organic compounds, sulphur is estimated as

(a) $BaSO_4$ (b) $BaCl_2$

(c) $Ba_3(PO_4)_2$ (d) H_2SO_4

194. In organic compounds, phosphorus is estimated as

(a) $Mg_2P_2O_7$ (b) $Mg(NH_4)PO_4$

(c) $Mg_3(PO_4)_2$ (d) H_3PO_4

195. 0.5 g of an organic compound on Kjeldahl's analysis gave enough ammonia to just neutralize 10 cm^3 of 1 M H_2SO_4. The percentage of nitrogen in the compound is

(a) 56 (b) 28

(c) 42 (d) 14

196. 0.28 g of a nitrogenous compound was kjeldahlised to produce 0.17 g of NH_3. The percentage of nitrogen in the organic compound is

(a) 5 (b) 30

(c) 50 (d) 80

197. 0.216 g of an organic compound gave 28.8 cm^3 of moist nitrogen measured at 288 K and 732.7 mm pressure. The percentage of nitrogen in the substance is approximately (Aq. tension at 288 K = 12.7 mm)

(a) 15 (b) 20

(c) 25 (d) 30

198. 0.10 g of an organic compound containing phosphorous gave 0.222 g of $Mg_2P_2O_7$. Then the percentage of phosphorus in it is

(a) 62 (b) 6.2

(c) 31 (d) 13

199. The molecular mass of an organic compound which contains only one nitrogen atom can be

(a) 73 (b) 76

(c) 146 (d) 152

200. An organic compound having carbon, hydrogen and sulphur contains 4% of sulphur. The minimum molecular weight of the compound is

(a) 200 (b) 400

(c) 600 (d) 800

201. 0.46 g of an organic compound containing carbon, hydrogen and oxygen was heated strongly in a stream of N_2 gas. The gaseous mixture thus obtained was passed first over heated coke at 1373 K and then through a warm solution of iodine pentoxide when 1.27 g of I_2 was liberated. The percentage of oxygen present in the organic compound is

(a) 87.34 (b) 13.91

(c) 47.38 (d) 38.47

202. 0.50 g of an organic compound was Kjeldahlised and the NH_3 evolved was absorbed in 50 ml of 0.5 M H_2SO_4. The residual acid required 60 cm^3 of 0.5 M NaOH. The percentage of nitrogen in the organic compound is

(a) 14 (b) 28

(c) 56 (d) 42

203. 0.59 g of the silver salt of an organic acid (mol. wt. 210) on ignition gave 0.36 g of pure silver. The basicity of the acid is

(a) 1 (b) 2
(c) 3 (d) 4

204. 0.532 g of the chloroplatinate of an organic base (mol. wt. 244) gave 0.195 g of Pt on ignition. Then the number of nitrogen atoms per molecule of the base is

(a) 1 (b) 2
(c) 3 (d) 4

205. A hydrocarbon having empirical formula as CH_2 has a density of 1.25 g/L. The molecular formula of the hydrocarbon is

(a) C_2H_2 (b) C_2H_4
(c) C_2H_6 (d) C_3H_8

206. The percentage composition of a compound is C = 90% and H = 10%. A plausible formula of the compound is

(a) C_8H_{10} (b) $C_{15}H_{30}$
(c) $C_{15}H_{20}$ (d) $C_{15}H_{12}$

207. Complete combustion of a sample of a hydrocarbon gives 0.66 g of CO_2 and 0.36 g of H_2O. The empirical formula of the compound is

(a) CH_2 (b) C_3H_4
(c) C_3H_8 (d) C_6H_8

208. A dibasic acid containing C, H and O was found to contain, C = 26.7% and H = 2.2%. The vapour density of its diethyl ester was found to be 73. What is the molecular formula of the acid ?

(a) CH_2O_2 (b) $C_2H_2O_4$
(c) $C_3H_3O_4$ (d) $C_4H_4O_4$

EXERCISE – 2

(Basic Level Subjective Questions)

BIOMOLECULES

1. What happens when ?
(i) Glucose is heated with phenyl hydrazine.
(ii) Glucose is treated with hydroxylamine.
(iii) Cane sugar is boiled with dilute sulphuric acid.
(iv) Cane sugar is warmed with dilute hydrochloric acid.
(v) Glucose is made to react with Tollen's reagent.
(vi) Glucose reacts with bromine water.
(vii) Glucose solution is heated with Fehling's solution.
(viii) Glucose is heated with conc. HNO_3.
(ix) Glucose is treated with HI in the presence of red phosphorus.
(x) Alanine is treated with nitrous acid.

2. (i) Classify the following into monosaccharides, disaccharides and polysaccharides :
Ribose, glycogen, maltose, deoxyribose, lactose, fructose, glucose, cane sugar, starch, cellulose.
(ii) Give one example of each of the following :
(a) Reducing sugar (b) Non-reducing sugar
(iii) Name the monomers of starch.
(iv) Name the monomers of cellulose.
(v) How many naturally occurring amino acids are there ?
(vi) Name the sugars present in nucleic acids.
(vii) Name the base present in RNA.
(viii) Name the disease caused due to the deficiency of
(a) Vitamin B_{12} (b) Vitamin C (c) Vitamin B_1
(ix) Name the protein that stores oxygen in the muscle tissue.

3. Answer the following :
(i) What are anomers ?
(ii) What are proteins ?
(iii) What is mutation ?
(iv) What is glucosidic linkage ?
(v) What is mutarotation ?
(vi) What is a peptide bond ?
(vii) What is denaturation of proteins ?
(viii) What is a nucleoside ? When does it become nucleotide ?
(ix) What is glycolysis ?
(x) What is digestion ?

4. Write short notes on
(i) Inversion of cane sugar.
(ii) Glycoside linkage.
(iii) Lipids.
(iv) Vitamins.
(v) Zwitter ion.
(vi) Isoelectric point.
(vii) Peptide linkage.
(viii) Coloured tests of proteins.
(ix) Mutarotation.
(x) Food digestion.

5. (a) Name the vitamins, deficiency of which cause :
(i) Night-blindness (ii) Rickets
(iii) Poor coagulation (iv) Beri-beri and
(b) Name the sources of following vitamins :
(i) Vitamin A (ii) Vitamin C
(iii) Vitamin D

POLYMERS

6. Write the names and structures of the monomers of following polymers
(i) Natural rubber (ii) Synthetic rubber
(iii) Nylon-66 (iv) Polythene
(v) PVC (vi) Teflon
(vii) Terylene (viii) Bakelite
(ix) Polypropylene (x) Neoprene
(xi) Plexiglass

7. How will you synthesise
(a) Polyvinyl chloride (PVC) from acetylene.
(b) Orlon from acetylene.
(c) Polystyrene from benzene.
(d) Terylene from ethylene and p-xylene.
(e) Nylon-66 from buta-1,3-diene.

8. Describe how the following polymers are synthesised ?
(i) Nylon-66 (ii) Thiokol
(iii) Buna-S (iv) Bakelite
(v) Terylene (vi) Melamine
(vii) Nylon-6

9. Write short notes on the following :
(a) Co-polymer (b) Synthetic rubber
(c) Thermoplastic (d) Thermosetting
(e) Elastomer (f) Vulcanization of rubber

CHEMISTRY IN EVERYDAY LIFE

10. Write short notes on the following :
(a) Chromogen (b) Auxochromes
(c) Ingrain dyes (d) Azo dyes
(e) Disperse dyes (f) Mordant dyes
(g) Vat dyes (h) Basic dyes

11. Answer the following :
(i) Name a drug in case of mental depression.
(ii) What are antacids ?
(iii) Write the name of chemicals used in food. Give their uses.
(iv) Give four examples of anti-histamines.
(v) Name the medicine which can act both as an analgesic as well as an antipyretic.
(vi) Name the main species responsible for malaria.
(vii) Which alkaloid is used for
(a) hypertension (b) malaria fever

12. Distinguish between :
(i) Antipyretics and antiseptics.
(ii) Antiseptics and disinfectants.
(iii) Broad spectrum and narrow spectrum antibiotics.
(iv) General and local anaesthetics.
(v) Narcotics and hypnotics.
(vi) Antioxidant and antimicrobial preservatives.

ENVIRONMENTAL CHEMISTRY

13. What is smog ? How is classical smog different from photochemical smogs ?

14. What are the reactions involved for ozone layer depletion in the stratosphere ?

15. What do you mean by Biochemical Oxygen Demand (BOD) ?

16. What do you mean by green chemistry ? How will it help decrease environmental pollution ?

QUALITATIVE AND QUANTITATIVE ANALYSIS

17. Will CCl_4 give white precipitate of AgCl on heating it with silver nitrate ? Give reason for your answer.

18. Why is a solution of potassium hydroxide used to absorb carbon dioxide evolved during the estimation of carbon present in an organic compound ?

19. Why is it necessary to use acetic acid and not sulphuric acid for acidification of sodium extract for testing sulphur by lead acetate test ?

20. A sample of 0.50 g of an organic compound was treated according to Kjeldahl's method. The ammonia evolved was absorbed in 50 ml of 0.5 M H_2SO_4. The residual acid required 60 mL of 0.5 M solution of NaOH for neutralisation. Find the percentage composition of nitrogen in the compound.

21. 0.45 g of an organic compound on combustion gave 0.729 g of carbon dioxide and 0.324 g water. Calculate the percentage of carbon and hydrogen in the compound.

22. 0.25 g of an organic compound when analysed by Duma's method yields 31.0 mL of nitrogen at STP. Determine the percentage of nitrogen in the compound.

23. 0.70 g of an organic compound was Kjeldahlised and the liberated ammonia was absorbed in 100 mL of N/10 H_2SO_4. After absorption the left over acid required 10 mL of N/5 NaOH for complete neutralisation. Calculate the percentage of nitrogen in the given compound.

24. 0.197 g of an organic substance when heated with excess of strong acid and silver nitrate gave 0.3525 g of silver iodide. Find the percentage of iodine in the compound.

25. 0.259 g of an organic compound containing sulphur was heated in a Carius tube with fuming nitric acid. The resultant was treated with barium chloride when 0.350 g of dry $BaSO_4$ was obtained. Calculate the percentage of sulphur in the given compound.

EXERCISE – 3

(JEE ADVANCED & JEE MAINS QUESTION)

BIOMOLECULES

JEE ADVANCED QUESTIONS

Single Answer Questions

1. Which of the following pairs give positive Tollen's test ? **(2004)**

(a) Glucose, sucrose

(b) Glucose, fructose

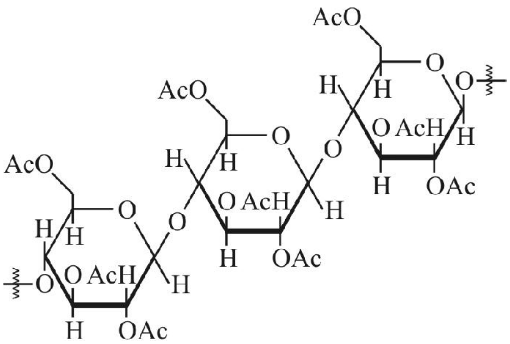
(c) Hexanal, acetophenone (d) Fructose, sucrose

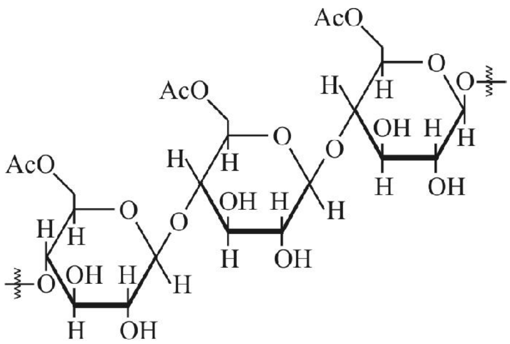
2. Two forms of D-glucopyranose, are called **(2005)**

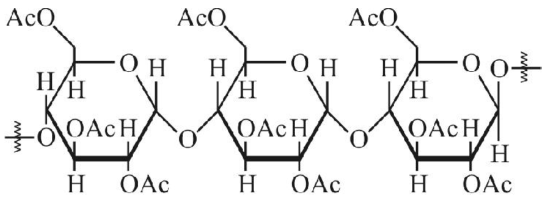
(a) enantiomers (b) anomers

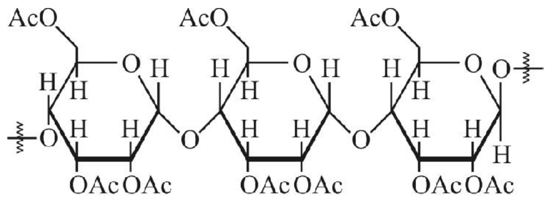
(c) epimers (d) diastereomers

3. Cellulose upon acetylation with excess acetic anhydride/H_2SO_4 (catalytic) gives cellulose triacetate whose structure is **(2008)**

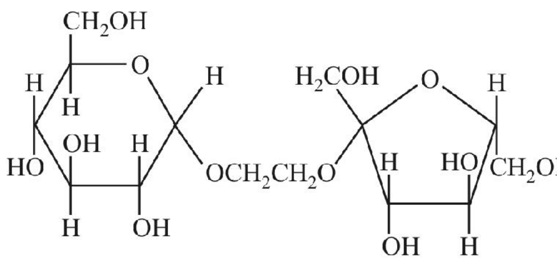
(a) 

(b) 

(c) 

(d) 

4. The correct statement about the following disaccharide is **(2018)**



(a) (b)

(a) Ring (a) is pyranose with α-glycosidic link

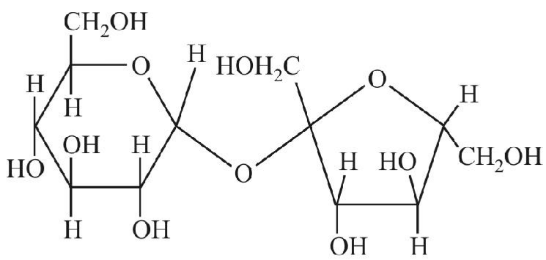
(b) Ring (a) is furanose with α-glycosidic link

(c) Ring (b) is furanose with α-glycosidic link

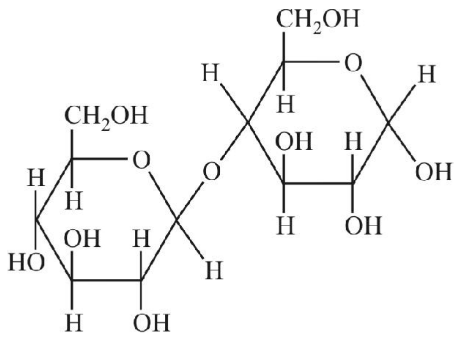
(d) Ring (b) is pyranose with β-glycosidic link

Multiple Answer Questions

5. The correct statement(s) about the following sugars X and Y is (are) **(2009)**



X



Y

(a) X is a reducing sugar and Y is a non-reducing sugar

(b) X is a non-reducing sugar and Y is a reducing sugar

(c) The glucosidic linkages in X and Y are α and β, respectively

(d) The glucosidic linkages in X and Y are β and α, respectively

Instructions : Q. No. 6 consists of an "ASSERTION" and a "REASON" use the following key to choose appropriate answer.

(A) If both ASSERTION and REASON are true and reason is the correct explanation of the assertion.

(B) If both ASSERTION and REASON are true but reason is not the correct explanation of the assertion.

(C) If ASSERTION is true but REASON is false.

(D) If both ASSERTION and REASON are false.

(E) If ASSERTION is false but REASON is true.

6. **Assertion** : Glucose gives a reddish-brown precipitate with Fehling's solution.

   Reason : Reaction of glucose with Fehling's solution gives CuO and gluconic acid. **(2007)**

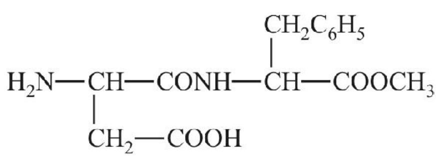
Subjective Questions

7. Write the structure of alanine at pH = 2 and pH = 10. **(2000)**

8. Give the structures of the products in the following reaction

   $$\text{Sucrose} \xrightarrow{H^{\oplus}} A + B$$ **(2000)**

9. Aspartame, an artificial sweetener, is a peptide and has the following structure **(2001)**



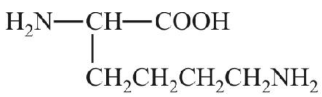
(i) Identify the four functional groups.

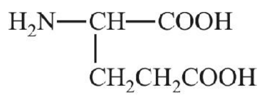
(ii) Write the Zwitter ionic structure.

(iii) Write the structures of the amino acids obtained from the hydrolysis of aspartame.

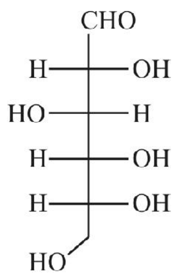
(iv) Which of the two amino acids is more hydrophobic ?

10. Following two amino acids lysine and glutamine form dipeptide linkage. What are two possible dipeptides ? **(2003)**





11. The structure of D-glucose is as follows : **(2004)**

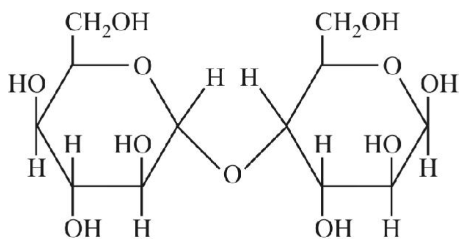


(a) Draw the structure of L-glucose.

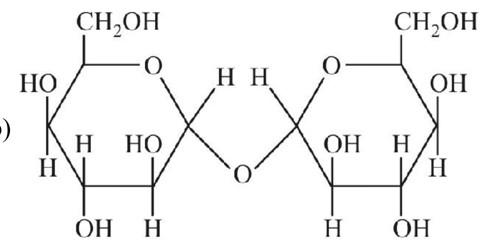
(b) Give the reaction of L-glucose with Tollen's reagent.

12. Which of the following disaccharide will not reduce Tollen's reagent ? **(2005)**

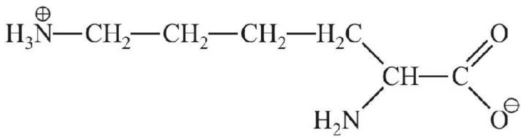
(a)



(b)



13. The total number of basic groups in the following form of lysine is



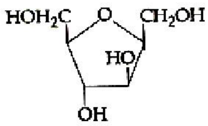
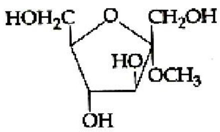
JEE MAINS QUESTION

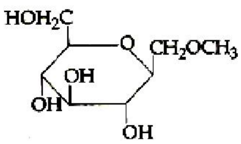
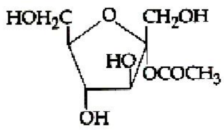
14. Glucose on prolonged heating with HI gives: **(2018)**

(a) 6-iodohexanal (b) n-Hexane

(c) l-Hexene (d) Hexanoic acid

15. Which of the following compounds will behave as a reducing sugar in an aqueous KOH solution ? **(2017)**

(a)  (b) 

(c)  (d) 

16. Thiol group is present in **(2016)**

(a) cystine (b) cysteine

(c) methionine (d) cytosine

17. Which of the following compounds can be detected by Molisch's test ? **(2014)**

(a) Nitro compounds (b) Sugars

(c) Amines (d) Primary alcohols

18. Which one of the following bases is not present in DNA ?

(a) Adenine (b) Cytosine **(2013)**

(c) Thymine (d) Quinoline

19. α-D(+)-glucose and β-D-(+)-glucose are **(2008)**

(a) conformers

(b) epimers

(c) anomers

(d) enantiomers

20. The secondary structure of a protein refers to **(2007)**

(a) α-helical backbone

(b) hydrophobic interactions

(c) sequence of α-amino acids

(d) fixed configuration of the polypeptide backbone

21. The term anomers of glucose refers to **(2006)**

(a) isomers of glucose that differ in configurations at carbons one and four (C-1 and C-4)

(b) a mixture of (D)-glucose and (L)-glucose

(c) enantiomers of glucose

(d) isomers of glucose that differ in configuration at carbon one (C-1)

22. The pyridimine bases present in DNA are **(2006)**

(a) cytosine and adenine

(b) cytosine and guanine

(c) cytosine and thymine

(d) cytosine and uracil

23. In both DNA and RNA, heterocylic base and phosphate ester linkages are at **(2005)**

(a) C'_5 and C'_1 respectively of the sugar molecule

(b) C'_1 and C'_5 respectively of the sugar molecule

(c) C'_2 and C'_5 respectively of the sugar molecule

(d) C'_5 and C'_2 respectively of the sugar molecule

24. Identify the correct statement regarding enzymes **(2004)**

(a) Enzymes are specific biological catalysts that can normally function at very high temperatures (T ~ 1000 K)

(b) Enzymes are normally heterogeneous catalysts that are very specific in their action

(c) Enzymes are specific biological catalysts that cannot be poisoned

(d) Enzymes are specific biological catalysts that possess well defined active sites

25. Insulin production and its action in human body are responsible for the level of diabetes. This compound belongs to which of the following categories ? **(2004)**

(a) A coenzyme

(b) A hormone

(c) An enzyme

(d) An antibiotic

26. Which base is present in RNA but not in DNA ? **(2004)**

(a) Uracil

(b) Cytosine

(c) Guanine

(d) Thymine

27. The reason for double helical structure of DNA is operation of **(2003)**

(a) van der Waals' forces

(b) dipole-dipole interaction

(c) hydrogen bonding

(d) electrostatic attractions

28. Complete hydrolysis of cellulose gives **(2003)**

(a) D-fructose (b) D-ribose

(c) D-glucose (d) L-glucose

29. A substance forms Zwitter ion. It can have functional groups **(2002)**

(a) $-NH_2, -COOH$

(b) $-NH_2, -SO_3H$

(c) Both (a) and (b)

(d) None of these

30. RNA contains **(2002)**

(a) ribose sugar and thymine

(b) ribose sugar and uracil

(c) deoxyribose sugar and uracil

(d) deoxyribose sugar and thymine

POLYMERS

JEE ADVANCED QUESTION

Single Answer Questions

31. Among cellulose, poly (vinyl chloride), nylon and natural rubber, the polymer in which the intermolecular force of attraction is weakest is **(2009)**

(a) nylon (b) poly (vinyl chloride)

(c) cellulose (d) natural rubber

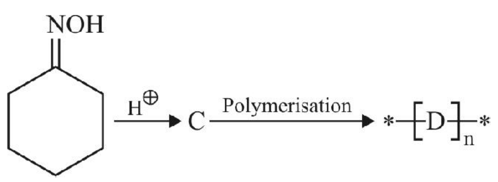
Match the Column

32. Match the chemical substances in Column-I with type of polymers/type of bond in Column-II. **(2007)**

Column-I	Column-II
(A) Cellulose	(P) Natural polymer
(B) Nylon-66	(Q) Synthetic polymer
(C) Protein	(R) Amide linkage
(D) Sucrose	(S) Glycoside linkage

Subjective Questions

33. Give the structures of the products in the following reaction



(2000)

34. Name the heterogenous catalyst used in the polymerization of ethylene. **(2003)**

JEE MAINS QUESTION

35. The formation of which of the following polymers involves hydrolysis reaction ? **(2017)**

(a) Nylon-6 (b) Bakelite

(c) Nylon-6,6 (d) Terylene

36. Which one is classified as a condensation polymer ? **(2014)**

(a) Dacron (b) Neoprene

(c) Teflon (d) Acrylonitrile

37. The polymer containing strong intermolecular forces, i.e. hydrogen bonding, is **(2011)**

(a) teflon (b) nylon-6,6

(c) polystyrene (d) natural rubber

38. Which of the following statements about low density polythene is FALSE ?

(a) It is a poor conductor of electricity.

(b) Its synthesis dioxygen or a peroxide initiator as a catalyst

(c) It is used in the manufacture of buckets, dust-bins etc.

(d) Its synthesis requires high pressure.

39. Bakelite is obtained from phenol by reacting with **(2008)**

(a) $(CH_2OH)_2$ (b) CH_3CHO

(c) CH_3COCH_3 (d) HCHO

40. Which of the following is fully fluorinated polymer ? **(2005)**

(a) PVC (b) Thiokol

(c) Teflon (d) Neoprene

41. Which of the following is a polyamide ? **(2005)**
(a) Bakelite (b) Terylene
(c) Nylon-66 (d) Teflon

42. Nylon threads are made up of **(2003)**
(a) polyvinyl polymer (b) polyester polymer
(c) polyamide polymer (d) polyethylene polymer

43. Monomers are converted to polymer by **(2002)**
(a) hydrolysis of monomers
(b) condensation reaction between monomers
(c) protonation of monomers
(d) none of these

CHEMISTRY IN EVERYDAY LIFE

JEE MAINS QUESTIONS

44. Which one of the following types of drugs reduces fever ? **(2005)**
(a) Tranquiliser (b) Antibiotic
(c) Antipyretic (d) Analgesic

45. Which of the following could act as a propellant for rockets ? **(2003)**
(a) Liquid hydrogen + liquid nitrogen
(b) Liquid oxygen + liquid argon
(c) Liquid hydrogen + liquid oxygen
(d) Liquid nitrogen + liquid oxygen

46. The distillation technique most suited for separating glycerol from spent-lye in the soap industry is : **(2016)**
(a) Fractional distillation (b) Steam distillation
(c) Distillation under reduced pressure
(d) Simple distillation

47. Which of the following is an anionic detergent ? **(2016)**
(a) Sodium lauryl sulphate
(b) Cetyltrimethyl ammonium bromide
(c) Glyceryl oleate.
(d) Sodium stearate

48. The recommended concentration of fluoride ion in drinking water is up to 1 ppm as fluoride ion is required to make teeth enamel harder by converting $[3Ca_3(PO_4)_2.Ca(OH)_2]$ to: **(2018)**
(a) $[3\{Ca(OH)_2\}.CaF_2]$ (b) $[CaF_2]$
(c) $[3(CaF_2).Ca(OH)_2]$ (d) $[3Ca_3(PO_4)_2.CaF_2]$

BIOMOLECULES, POLYMERS CHEMISTRY IN EVERYDAY LIFE

JEE MAINS ONLINE QUESTION

BIOMOLECULES

1. Which of the following will not shown mutarotation? **Online 2014 SET (3)**
(a) Lactose (b) Sucrose
(c) Glucose (d) Maltose

2. The reason for double helical structure of DNA is the operation of : **Online 2014 SET (4)**
(a) Electrostatic attraction
(b) Dipole – Dipole interactions
(c) Hydrogen bonding
(d) van der Waals forces

3. Accumulation of which of the following molecules in the muscles occurs as a result of vigorous exercise ? **Online 2015 SET (1)**
(a) Glycogen (b) Glucose
(c) Pyruvic acid (d) L-lactic acid

4. The two forms of D-glucopyranose obtained from the solution of D-glucose are called **Online 2016 SET (1)**

(a) Isomer (b) Anomer

(c) Epimer (d) Enantiomer

5. Observation of "Rhumann's purple" is a confirmatory test for the presence of : **Online 2016 SET (2)**

(a) Reducing sugar (b) Cupric ion

(c) Protein (d) Starch

6. An antibiotic contains nitro group attached to aromatic nucleus in its structure. It is **Online 2017 SET (1)**

(a) pencillin (b) streptomycin

(c) tetracyclin (d) chloramphenicol

7. Among the following, the essential amino acid is :

Online 2017 SET (1)

(a) Alanine (b) Valine

(c) Aspartic acid (d) Serine

8. The incorrect statement among the following is :

Online 2017 SET (2)

(a) α-D-glucose and β-D-glucose are anomers.

(b) α-D-glucose and β-D-glucose are enantiomers.

(c) Cellulose is a straight chain polysaccharide made up of only β-D-glucose units.

(d) The penta acetate of glucose does not react with hydroxyl amine.

9. Among the following, the incorrect statement is :

Online 2018 SET (3)

(a) Maltose and lactose has 1, 4-glycosidic linkage

(b) Sucrose and amylose has 1, 2-glycosidic linkage.

(c) Cellulose and amylose has 1, 4-glycosidic linkage.

(d) Lactose contains β -D-galactose and β -D-glucose.

10. Fructose and glucose can be distinguished by :

Online 2019 SET (1)

(a) Benedict's test (b) Fehling's test

(c) Barfoed's test (d) Seliwanoff's test

11. Which of the following statements is not true about sucrose ? **Online 2019 SET (2)**

(a) The glycosidic linkage is present between C_1 of α-glucose and C_1 of β-fructose

(b) It is a non-reducing sugar

(c) It is also named as invert sugar

(d) On hydrolysis it produces glucose and fructose

POLYMER

12. Which one of the following class of compounds is obtained by polymerization of acetylene?

Online 2014 SET (1)

(a) Poly-yne (b) Poly-ene

(c) Poly-ester (d) Poly-amide

13. Which one of the following is an example of thermosetting polymers ? **Online 2014 SET (4)**

(a) Nylon 6,6 (b)Bakelite

(c) Neoprene (d) Buna–N

14. Which of the following polymers is synthesized using a free radical polymerization technique ?

Online 2016 SET (2)

(a) Teflon (b) Terylene

(c) Melamine polymer (d) Nylon 6, 6

15. Which of the following statements is not true ?

Online 2018 SET (2)

(a) Step growth polymerisation requires a bifunctional monomer

(b) Nylon 6 is an example of step growth polymerisation

(c) Chain growth polymerisation includes both homopolymerisation and copolymerisation

(d) Chain growth polymerisation involves homopolymerisation only

16. Which of the following is a condensation polymer ?

Online 2019 SET (2)

(a) Nylon 6, 6 (b) Neoprene

(c) Buna-S (d) Teflon

17. Which of the following is a thermosetting polymer ?

Online 2019 SET (3)

(a) PVC (b) Bakelite

(c) Buna-N (d) Nylon 6

CHEMISTRY IN EVERYDAY LIFE

18. Which one of the following is used as Antihistamine?

Online 2014 SET (2)

(a) Norethindrone (b) Diphenhydramine

(c) Chloranphenicol (d) Omeprazole

19. Aminoglycosides are usually used as:

Online 2014 SET (3)

(a) analgesic (b) hypnotic

(c) antibiotic (d) antifertility

20. Under ambient conditions, which among the following surfactants will form micelles in aqueous solution at lowest molar concentration ? **Online 2015 SET (1)**

(a) $CH_3-(CH_2)_8-COO^- Na^+$

(b) $CH_3(CH_2)_{11}\overset{\oplus}{N}(CH_3)_3 Br^-$

(c) $CH_3-(CH_2)_{13}-OSO_3^- Na^+$

(d) $CH_3(CH_2)_{15}\overset{\oplus}{N}(CH_3)_3 Br^-$

21. The artificial sweetener that has the highest sweetness value in comparison to cane sugar is :

Online 2016 SET (1)

(a) Aspartane (b) Saccharin

(c) Sucralose (d) Alitame

22. Which one of the following substances used in dry cleaning is a better strategy to control environmental pollution ? **Online 2016 SET (2)**

(a) Tetrachloroethylene (b) Carbon dioxide

(c) Sulphur dioxide (d) Nitrogen dioxide

23. Which of the following is a bactericidal antibiotic ?

Online 2018 SET (3)

(a) Erythromycin (b) Tetracycline

(c) Chloramphenicol (d) Ofloxacin

24. Noradrenaline is a/an **Online 2019 SET (2)**

(a) Neurotransmitter (b) Antidepressant

(c) Antihistamine (d) Antacid

ENVIRONMENTAL CHEMISTRY

25. Which of the following statements about the depletion of ozone layer is correct? **Online 2014 SET (2)**

(a) The problem of ozone depletion is less serious at poles because NO_2 solidifies and is not available for consuming $ClO^\bullet$ radicals.

(b) The problem of ozone depletion is more serious at poles because ice crystals in the clouds over poles act as catalyst for photochemical reactions involving the decomposition of ozone by $Cl^\bullet$ and $ClO^\bullet$ radicals.

(c) Oxides of nitrogen also do not react with ozone in stratosphere.

(d) Freons, chlorofluorocarbons, are inert chemically, they do not react with ozone in stratosphere.

26. Global warming is due to increase of:

Online 2014 SET (3)

(a) methane and Co in atmosphere

(b)methane and nitrous oxide in atmosphere

(c) methane and O_3 in atmosphere

(d) methane and CO_2 in atmosphere

27. Addition of phosphate fertilisers to water bodies causes : **Online 2015 SET (1)**

(a) increase in amount of dissolved oxygen in water

(b) deposition of calcium phosphate

(c) increase in fish population

(d) enhanced growth of algae

28. BOD stands for : **Online 2016 SET (1)**

(a) Biological Oxygen Demand

(b) Bacterial Oxidation Demand

(c) Biochemical Oxygen Demand

(d) Biochemical Oxidation Demand

29. Identify the pollutant gases largely responsible for the discoloured & lustreless nature of marble of the Taj Mahal

Online 2017 SET (1)

(a) O_3 and CO_2 (b) CO_2 and NO_2

(c) SO_2 and NO_2 (d) SO_2 and O_3

30. Which of the following is a set of green house gases ?

Online 2017 SET (2)

(a) CH_4, O_3, N_2, SO_2

(b) O_3, N_2, CO_2, NO_2

(c) O_3, NO_2, SO_2, Cl_2

(d) CO_2, CH_4, N_2O, O_3

31. Biochemical Oxygen Demand (BOD) value can be a measure of water pollution caused by the organic matter. Which of the following statements is correct ?

(Online 2018 SET (2)

(a) Aerobic bacteria decrease the BOD value

(b) Anaerobic bacteria increase the BOD value

(c) Clean water has BOD value higher than 10 ppm

(d) Polluted water has BOD value higher than 10 ppm.

32. Which is wrong with respect to our responsibility as a human being to protect our environment ?

(Online 2019 SET (1)

(a) Restricting the use of vehicles

(b) Using plastic bags

(c) Setting up compost tin in gardens

(d) Avoiding the use of floodlighted facilities.

33. The maximum prescribed concentration of copper in drinking water is : **(Online 2019 SET (2)**

(a) 0.5 ppm (b) 3 ppm

(c) 5 ppm (d) 0.05 ppm

34. Excessive release of CO_2 into the atmosphere results in :

(Online 2019 SET (3)

(a) formation of smog (b) depletion of ozone

(c) global warming (d) polar vortex

ANSWER KEY

EXERCISE–1

1. (d)	**2.** (a)	**3.** (d)	**4.** (d)	**5.** (d)	**6.** (d)	**7.** (a)	**8.** (d)	**9.** (d)	**10.** (c)
11. (c)	**12.** (b)	**13.** (c)	**14.** (a)	**15.** (a)	**16.** (b)	**17.** (a)	**18.** (d)	**19.** (a)	**20.** (c)
21. (b)	**22.** (c)	**23.** (b)	**24.** (b)	**25.** (c)	**26.** (c)	**27.** (a)	**28.** (c)	**29.** (b)	**30.** (b)
31. (c)	**32.** (b)	**33.** (b)	**34.** (c)	**35.** (d)	**36.** (c)	**37.** (d)	**38.** (a)	**39.** (b)	**40.** (c)
41. (d)	**42.** (a)	**43.** (d)	**44.** (a)	**45.** (b)	**46.** (d)	**47.** (b)	**48.** (a)	**49.** (b)	**50.** (d)
51. (b)	**52.** (d)	**53.** (b)	**54.** (d)	**55.** (b)	**56.** (b)	**57.** (a)	**58.** (b)	**59.** (c)	**60.** (d)
61. (c)	**62.** (d)	**63.** (c)	**64.** (b)	**65.** (c)	**66.** (a)	**67.** (b)	**68.** (a)	**69.** (b)	**70.** (b)
71. (c)	**72.** (d)	**73.** (d)	**74.** (a)	**75.** (a)	**76.** (d)	**77.** (d)	**78.** (b)	**79.** (a)	**80.** (d)
81. (c)	**82.** (b)	**83.** (b)	**84.** (d)	**85.** (c)	**86.** (d)	**87.** (b)	**88.** (c)	**89.** (b)	**90.** (b)
91. (a)	**92.** (d)	**93.** (d)	**94.** (d)	**95.** (b)	**96.** (c)	**97.** (d)	**98.** (d)	**99.** (b)	**100.** (a)
101. (d)	**102.** (c)	**103.** (d)	**104.** (b)	**105.** (d)	**106.** (b)	**107.** (c)	**108.** (c)	**109.** (c)	**110.** (c)
111. (d)	**112.** (d)	**113.** (a)	**114.** (d)	**115.** (b)	**116.** (c)	**117.** (b)	**118.** (d)	**119.** (a)	**120.** (d)
121. (c)	**122.** (a)	**123.** (b)	**124.** (d)	**125.** (a, b, c, d)		**126.** (a)	**127.** (a)	**128.** (c)	
129. (d)	**130.** (a)	**131.** (b)	**132.** (b)	**133.** (a)	**134.** (b)	**135.** (c)	**136.** (c)	**137.** (c)	**138.** (b)
139. (d)	**140.** (c)	**141.** (d)	**142.** (b)	**143.** (a)	**144.** (c)	**145.** (c)	**146.** (c)	**147.** (b)	**148.** (a)
149. (c)	**150.** (c)	**151.** (d)	**152.** (a)	**153.** (b)	**154.** (b)	**155.** (b)	**156.** (b)	**157.** (a)	**158.** (c)
159. (d)	**160.** (a)	**161.** (b)	**162.** (a)	**163.** (b)	**164.** (c)	**165.** (d)	**166.** (a)	**167.** (a)	**168.** (c)
169. (d)	**170.** (d)	**171.** (a)	**172.** (d)	**173.** (d)	**174.** (d)	**175.** (d)	**176.** (a)	**177.** (a)	**178.** (a)
179. (c)	**180.** (d)	**181.** (c)	**182.** (a)	**183.** (a)	**184.** (a)	**185.** (b)	**186.** (a)	**187.** (d)	**188.** (d)
189. (b)	**190.** (a)	**191.** (a)	**192.** (d)	**193.** (a)	**194.** (a)	**195.** (a)	**196.** (c)	**197.** (a)	**198.** (a)
199. (a)	**200.** (d)	**201.** (b)	**202.** (c)	**203.** (c)	**204.** (d)	**205.** (b)	**206.** (c)	**207.** (c)	**208.** (b)

EXERCISE – 2

1. (i) $$CH_2OH(CHOH)_4CHO + H_2N.NHC_6H_5 \longrightarrow \underset{\text{Glucose phenylhydrazone}}{CH_2OH(CHOH)_4CH{=}NNHC_6H_5}$$

With excess of phenylhydrazine, glucosazone, $\begin{array}{l} CH_2OH(CHOH)_3{-}C{=}NNHC_6H_5 \\ \qquad\qquad\qquad\quad\ \, | \\ \qquad\qquad\qquad\quad CH{=}NNHC_6H_5 \end{array}$, is formed.

(ii) $$CH_2OH(CHOH)_4CHO + H_2NOH \longrightarrow \underset{\text{Glucose oxime}}{CH_2OH.(CHOH)_4CH{=}NOH}$$

(iii) $$\underset{\text{Cane sugar}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{\text{Dil. } H_2SO_4} \underset{\text{Glucose}}{C_6H_{12}O_6} + \underset{\text{Fructose}}{C_6H_{12}O_6}$$

(iv) Same products as in (iii)

(v) $$C_6H_{12}O_6 + 2Ag(NH_3)_2OH \longrightarrow \begin{array}{c} COONH_4 \\ | \\ (CHOH)_4 \\ | \\ CH_2OH \end{array} + \underset{\text{Silver mirror}}{2Ag} + 3NH_3 + H_2O$$

(vi) $$C_6H_{12}O_6 + [O] \xrightarrow{Br_2/H_2O} \underset{\text{Gluconic acid}}{C_6H_{12}O_7}$$

(vii) $C_6H_{12}O_6 + 2Cu(OH)_2 + NaOH \longrightarrow CH_2OH(CHOH)_4COONa + \underset{\text{Red ppt.}}{Cu_2O} + 3H_2O$

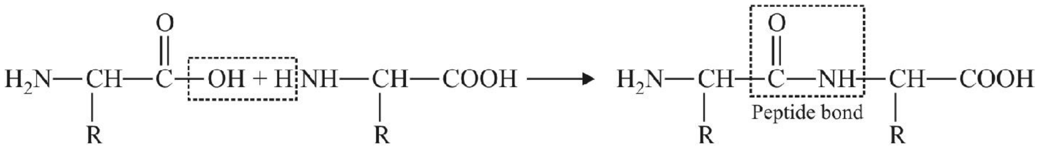
(viii) $CH_2OH(CHOH)_4CHO + 2[O] \xrightarrow{HNO_3} \underset{\text{Glucaric acid}}{COOH.(CHOH)_4COOH}$

(ix) n-Hexane is formed.

(x) $CH_3\underset{NH_2}{\underset{|}{C}}HCOOH + HONO \longrightarrow CH_3\underset{OH}{\underset{|}{C}}HCOOH + N_2 + H_2O$

2. (i) Monosaccharides : Ribose, deoxyribose, glucose, fructose.
Disaccharides : Maltose, lactose, cane sugar.
Polysaccharides : Glycogen, starch, cellulose.
(ii) Reducing sugar – Glucose; Non-reducing sugar–cane sugar (sucrose). (iii) α-Glucose.
(iv) β-Glucose. (v) Twenty. (vi) Deoxyribose and ribose. (vii) Adenine; cytosine; guanine; uracil.
(viii) (a) Pernicious anaemia, (b) Breakdown of immune defence system, (c) Beri-beri. (ix) Myoglobin.

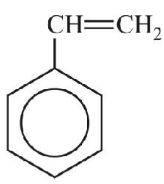
3. (i) A pair of isomers that differ with respect to orientation of the –OH group at carbon-1.
(ii) Proteins are polymers of amino acids having peptide linkage.
(iii) Sudden change in the sequence of nitrogenous bases along the DNA strand and can synthesise a new protein with altered amino acid sequence.
(iv) The linkage formed by the combination of hydroxyl group of the hemiacetal carbon of one monosaccharide with the –OH group of the other monosaccharide.
(v) The process that involves the change in the optical rotation of either form of glucose in aqueous solution to that of equilibrium mixture is known as mutarotation.
(vi) The linkage that unites various amino acid units in a peptide molecule is known as peptide bond.



(vii) On heating or on treatment with mineral acids globular proteins undergo coagulation to give fibrous protein. It loses whole or a part of biological activity.
(viii) The molecule in which one of the organic bases is linked with a sugar is called nucleoside. When a phosphate group is also attached to the nucleoside, the compound is known as nucleotide.
(ix) It is the anaerobic conversion of 6-carbon atoms compound, glucose, into two molecules of pyruvic acid.
(x) It is the process of breaking up of a complex food into simple molecules.

5. (a) (i) Vitamin A, (ii) Vitamin D, (iii) Vitamin K, (iv) Vitamin B_1
(b) (i) Synthesised in cells of liver and intestinal mucous membrane from carotenoid pigments found in milk, butter, egg yolk, liver, fish oil, etc.
(ii) Lemon, orange and other citrus fruits, tomatoes, green vegetables, etc.
(iii) Synthesised in skin cells in sunlight, found in butter, liver egg yolk, fish oil, etc.

6. (i) Isoprene $\left(CH_2{=}\overset{CH_3}{\overset{|}{C}}{-}CH{=}CH_2\right)$;

(ii) 1,3-Butadiene $(CH_2{=}CH{-}CH{=}CH_2)$ and styrene  ;

(iii) Hexamethylene diamine $[H_2N(CH_2)_6NH_2]$ and adipic acid $[HOOC(CH_2)_4COOH]$;
(iv) Ethylene $(CH_2{=}CH_2)$; (v) Vinyl chloride $(CH_2{=}CHCl)$;

(vi) Tetrafluoro ethylene ($CF_2{=}CF_2$); (vii) Ethylene glycol ($HOCH_2–CH_2OH$) and phthalic acid;

(viii) Phenol and formaldehyde; (ix) Propylene ($CH_3CH{=}CH_2$);

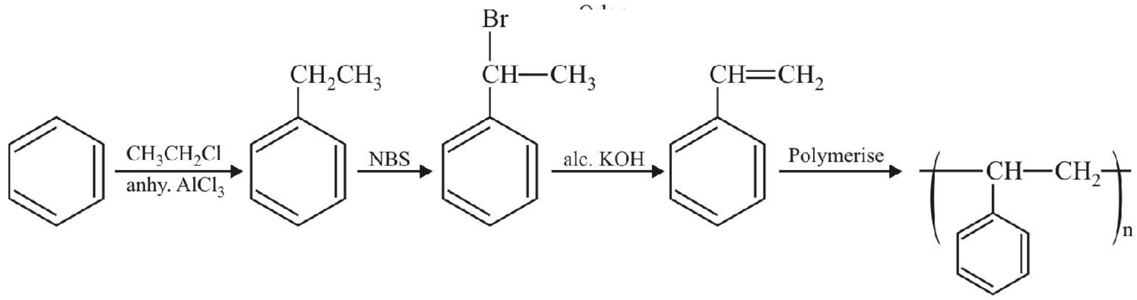
(x) Chloroprene ($CH_2{=}C(Cl){-}CH{=}CH_2$);

(xi) Methyl methylacrylate ($CH_2{=}C(CH_3){-}COOCH_3$)

7. (a) $nCH{\equiv}CH + HCl \longrightarrow nCH_2{=}CH(Cl) \xrightarrow{\text{Polymerise}} \left(\!-CH_2-CH(Cl)-\right)_n$

PVC

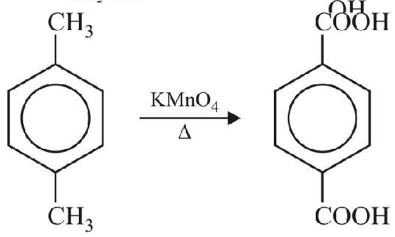
(b) $nCH{\equiv}CH \xrightarrow{HCN} nCH_2{=}CH(CN) \xrightarrow{\text{Polymerise}} \left(\!-CH_2-CH(CN)-\right)_n$

(c) 

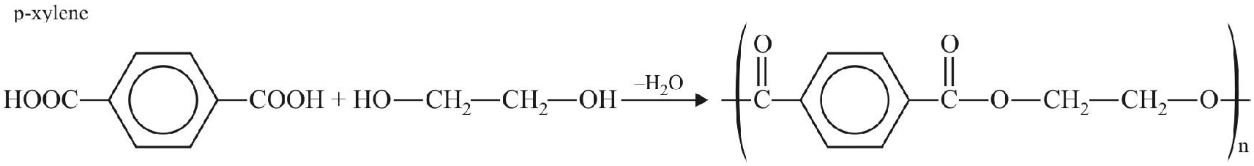
Polystyrene

(d) $CH_2{=}CH_2 \xrightarrow{\text{alk. } KMnO_4} CH_2(OH)-CH_2(OH)$

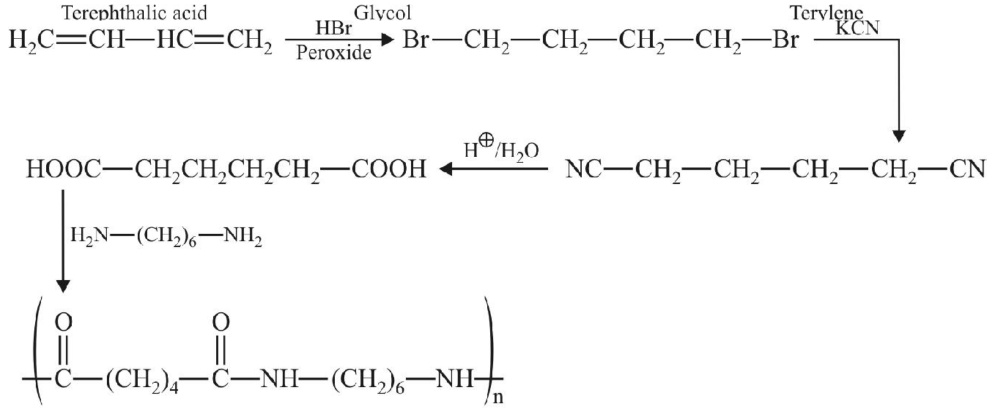
Ethylene



p-xylene



Terephthalic acid Glycol Terylene

(e) 

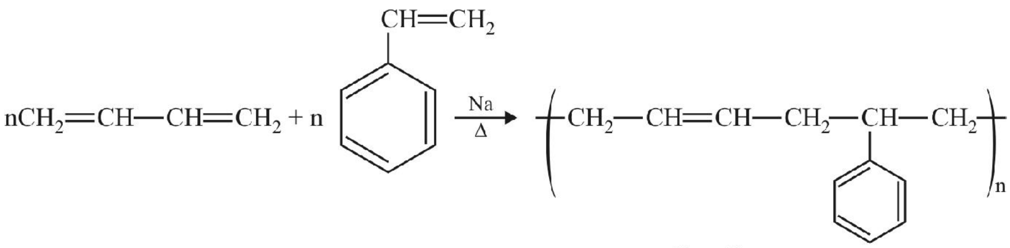
Nylon-66

8. (i) $$n\,HO-\underset{\underset{O}{\|}}{C}-(CH_2)_4-\underset{\underset{O}{\|}}{C}-OH + n\,H_2N-(CH_2)_6-NH_2 \xrightarrow[-H_2O]{} \left(\underset{\underset{O}{\|}}{C}-(CH_2)_4-\underset{\underset{O}{\|}}{C}-NH-(CH_2)_6-NH\right)_n$$

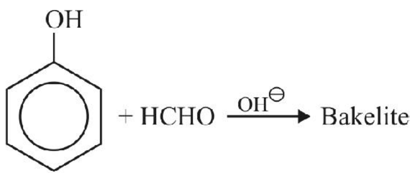
Adipic acid Hexamethylenediamine Nylon-66

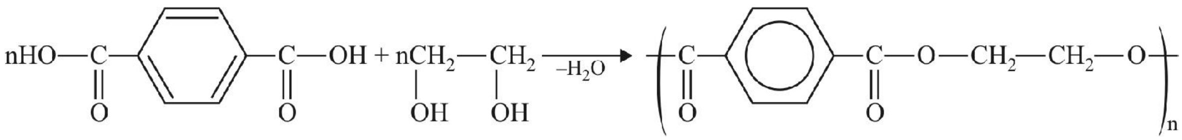
(ii) $$Cl-CH_2CH_2-Cl + Na-S-S-Na + Cl-CH_2-CH_2-Cl \xrightarrow{\text{Polymerise}} -CH_2-CH_2-S-S-CH_2-CH_2-$$

Thiokol

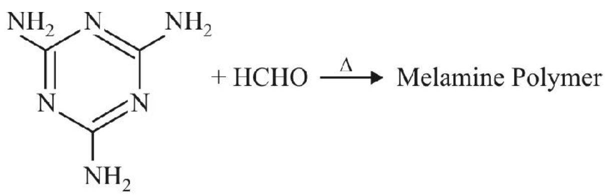
(iii) 

Buna-S

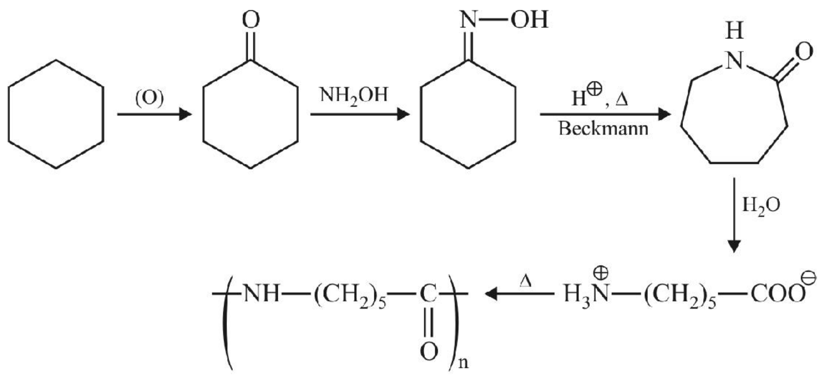
(iv) 

(v) 

Terylene

(vi) 

Melamine

(vii) 

Nylon-6

11. (i) Fluoxetine, Venlafaxine, fluvoxamine, citalopram etc.

(ii) Those substances which increases the pH of stomach fluids and removes the acidity e.g. milk of magnesia.

(iii) Antioxidants : To prevent oxidation of fats.

Artificial sweetening agents : To increase sweetness number of foods.

Preservatives : To protect food from fungi, yeast, bacteria etc.

Edible colours : To improve the appearance of foods.

(iv) Avomine, Antistine, Foristal, Acididil etc.

(v) Brufen

(vi) Plasmodium molariae

(vii) (a) Digitalis, Propronolol, (b) Quinine

14. (i) $Cl_2CF_2 \xrightarrow{hv} \dot{C}l + Cl\dot{C}F_2$

(ii) $\dot{C}l + O_3 \longrightarrow Cl\dot{O} + O_2$

(iii) $Cl\dot{O} + O \longrightarrow \dot{C}l + O_2$

17. No, because Cl atoms in CCl_4 are non-ionisable.

18. KOH is basic and CO_2 is an acidic oxide. So KOH absorbs CO_2 efficiently.

19. If we use H_2SO_4, $Pb(CH_3COO)_2$ will react with it to give white ppt. of $PbSO_4$ so we use acetic acid.

20. % of nitrogen = 56

21. C = 44%, H = 8%

22. N = 15.5%

23. N = 16%

24. I = 96.7%

25. S = 18.5%

EXERCISE – 3

1. (b)	**2.** (b)	**3.** (a)	**4.** (a)	**5.** (b, c)	**6.** (c)	**13.** 0002	**14.** (b)	**15.** (d)	**16.** (b)
17. (b)	**18.** **(d)**	**19.** (c)	**20.** (a)	**21.** (d)	**22.** (c)	**23.** (a)	**24.** (d)	**25.** (b)	**26.** (a)
27. (c)	**28.** (c)	**29.** (c)	**30.** (b)	**31.** (d)	**32.** (A) – P, S ; (B) – Q, R ; (C) – P, R ; (D) – S				
35. (a)	**36.** (a)	**37.** (b)	**38.** (c)	**39.** (d)	**40.** (c)	**41.** (c)	**42.** (c)	**43.** (b)	
44. (c)	**45.** (c)	**46.** (c)	**47.** (a)	**48.** (d)					

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1. (b)	**2.** (c)	**3.** (d)	**4.** (b)	**5.** (c)	**6.** (d)	**7.** (b)	**8.** (b)	**9.** (b)	**10.** (d)
11. (a)	**12.** (b)	**13.** (b)	**14.** (a)	**15.** (d)	**16.** (a)	**17.** (b)	**18.** (b)	**19.** (c)	**20.** (d)
21. (d)	**22.** (b)	**23.** (d)	**24.** (a)	**25.** (b)	**26.** (d)	**27.** (d)	**28.** (c)	**29.** (c)	**30.** (d)
31. (d)	**32.** (b)	**33.** (b)	**34.** (c)						

Dream on !!